Staples STEM Journal

Astronomy, AI, Stem Cells, Vaccines, Hemifusomes, CRISPR, Cancer, TOXBOX, Qbits, Comets

STAPLES HIGH SCHOOL WINTER 2025/26

The Staples STEM Journal provides an outlet for individuals to share their STEM interests with the Staples community and aims to broaden public interest and knowledge in these fields.

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We'd like to thank our excellent staff advisor, Mrs. Amy Parent, for her support and guidance

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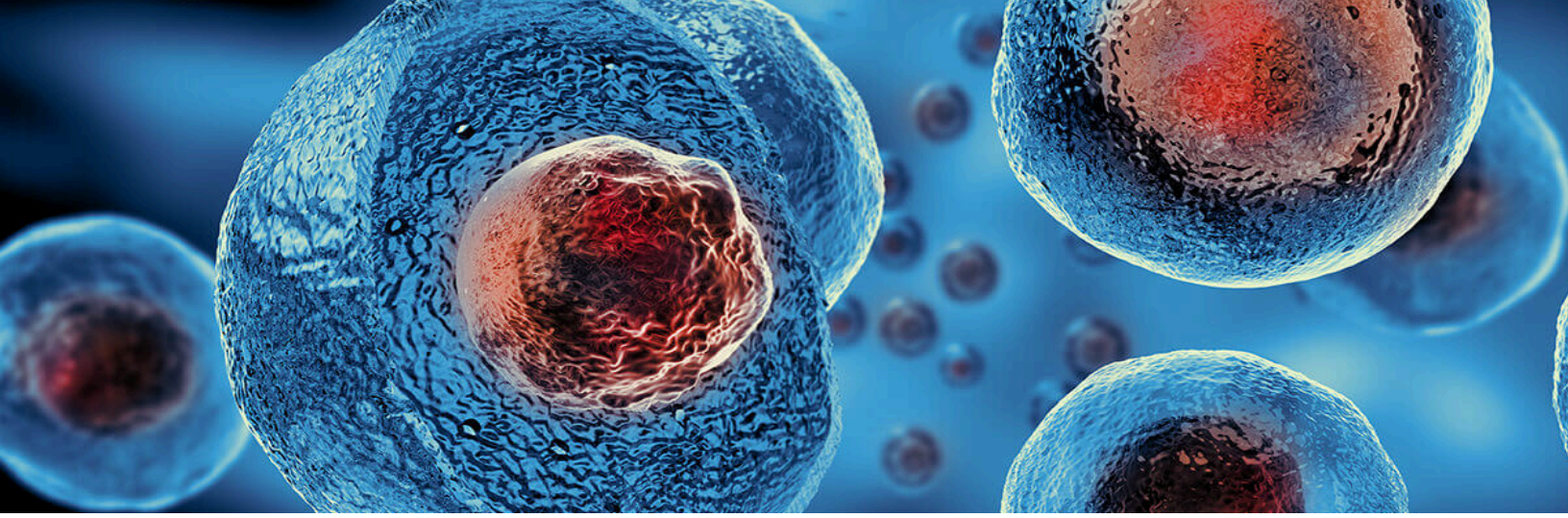
Letter from the Editor

Dear Reader,

I'm excited to share the fall/winter issue of the Staples STEM Journal, which is the culmination of many students' hard work and efforts throughout the first semester.
I would first like to take this opportunity to thank our members, who produced the fascinating articles published in this issue. I would also like to thank our wonderful leadership team, who made this issue not only possible but a great success. Finally, I would like to thank Ms. Amy Parent, our fantastic faculty advisor. Ms. Parent has devoted countless hours of her time to the success and improvement of the STEM Journal and has been the driving force behind ensuring that this publication has had the quality and professionalism that is expected of it.
Over the past year, we struggled significantly with funding, among several other obstacles standing in the way of our success as a club and as a publication. Individual members of the club at-large and of the leadership team worked together to overcome these challenges. The efforts of the past few months have ensured that the Staples STEM Journal, which provides a rare but truly important opportunity for students to combine their scientific and writing abilities, continues to thrive.
I believe that this latest issue showcases the research and writing skills of our student writers. This semester, we covered a wide range of topics, from the anti-aging effects of stem cells, to the discovery of a new interstellar object, to the creation of organoids. As a result, this is one of the most fascinating issues yet.
This semester, we had the privilege of hearing a Zoom lecture from Dr. Ron Weiss, an M.I.T. Professor of Biological Engineering, Principal Researcher at the Weiss Lab, and a pioneer in the field of synthetic biology. He explained to us his very interesting work, such as the fabrication of synthetic DNA circuits which operate much like a computer.
As we enter the spring, we are transitioning to new leadership. I am excited to welcome Sienna Schwartz as our new Editor-in-Chief. I would like to thank all of the outgoing members of the leadership team; Reia Bhardwaj, our Editor-in-Chief emeritus; Nolan Francis, our advertising coordinator; and Eric Wilson, a staff editor.
I am very excited to watch the STEM Journal continue to evolve over the next semester and in the coming years!

Enjoy our Fall/Winter 2025 issue!

Best wishes,
Gray McGuinness '26
Editor-in-Chief



How Regeneration and Longevity Breakthroughs Using Stem Cells Can Eventually Lead to Living Longer

Valentina Belyaev, '29

What Stem Cells Are, Why They Matter, and the Goal of Longevity

Imagine being able to reverse aging in the human body and repair damaged organs by using cells. Stem cells are cells that can regenerate and renew their properties to transform into different cells in areas of the body that need repair. They are the only type of cell found in organisms that contain the ability to generate new cell types and are technically genetically complete. An organism that is genetically

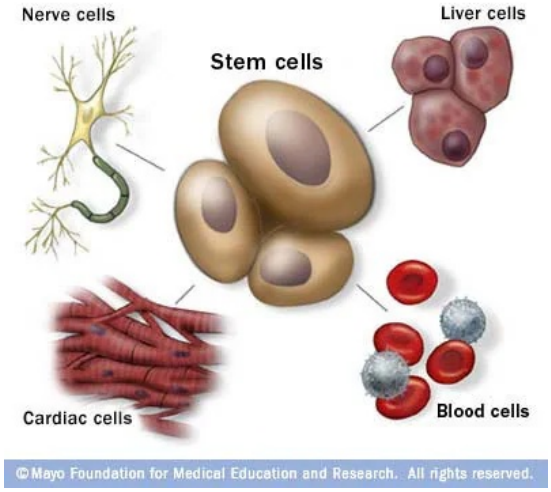


Figure 1: The types of cells that stem cells can transform into [Illustration]. (Mayo Clinic, 2024)

complete has a full set of genetic information. Stem cells contain the full genetic code that allows them to become any cell type. The types of cells that stem cells can transform into are shown in Figure 1. Once they are able to transform into other cells, that is when they become a more specialized type of cell. The goal of longevity by using stem cells is to extend an individual's life span and reverse aging related damage to the body. Stem cells matter because they can be used safely in procedures and they are something that the body already naturally produces. Some examples of cell therapy are bone marrow transplants and skin regeneration. They can easily be used for therapy to help improve the bodily functions of an organism, containing characteristics like engraftment, transplantation, and differentiation. Stem cells can also potentially answer the question why organisms can't live forever.

The Science Behind Stem Cells and Their Anti-Aging Effects

When researching stem cells, researchers prefer to use embryo stem cells rather than adult stem cells. But why is that? This is because adult stem cells are not as versatile and long-lasting compared to embryo stem cells. According to Mayo Clinic, "Adult stem cells are also more likely to contain irregularities due to environmental hazards, such as toxins, or from errors acquired by the cells during replication" (Mayo Clinic, 2024). Embryo cells are taken from embryos when they are typically 3 to 5 days old mainly because they are fresh and have about 150 cells in total. However, the use of embryonic stem cells raises ethical concerns, which is why options like induced pluripotent stem cells (iPSCs) are also considered for use in this process. This is mainly because molecular science is based on transcription and epigenetic mechanisms, which are the main factors that allow stem cells to become a specialized cell. "Signals for cell differentiation include factors secreted by other cells, physical contact with neighboring cells, and certain molecules in the microenvironment" (National Institutes of Health, 2016). Diseases that are associated with aging often involve the degeneration of cells, where the

structure of cells weaken and their purpose of functioning often becomes less useful and impactful to the body. Stem cell based therapy can help rejuvenate our bodies by replacing damaged cells or promoting repair. Stem cells are great for treating aging-related diseases or natural aging over time because they have a large capacity for self-renewal. By doing this, one may be younger and healthier, both internally and externally. This is because they slow cellular aging.

Worldwide Breakthroughs in Longevity Research

In 2025, a research study called "Senescence-resistant human mesenchymal progenitor cells counter aging in primates" was researched in China. This study was conducted by the Chinese Academy of Sciences (CAS) and Capital Medical University. They researched super stem cells, which have enhanced

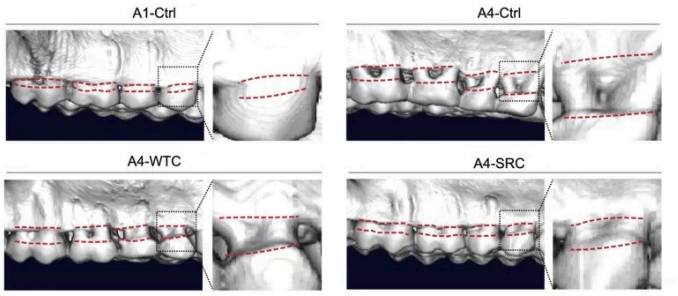


Figure 2: The differences of oral bone health in young monkeys and old monkeys with extreme bone loss. Old monkeys treated with stem cells had their bone health significantly improve [Image]. (NAD', 2025)

regeneration abilities compared to normal stem cells. Initially, the monkeys that they were testing this on had old and had damaged tissues in their body, suffering immune systems, and memory loss. After 44 weeks of biweekly injections, it was found that "'Super stem cells' improved the memory of monkeys while protecting against neurodegeneration, prevented age-related bone loss while rejuvenating over 50% of the 61 tissues analyzed, and reduced inflammation and senescent cells (cells that accumulate to promote aging)" (NAD+, 2025). In addition to this, the scientists found that the stem cells were able to fix and repair organs of the monkey's bodies, such as their bones. Before testing began, the monkeys had weak and brittle bones that were prone to breaking or fracturing. After the experiment, it was found that the monkeys that were given higher doses of stronger stem cells improved their oral health, making their teeth stronger and more similar to those of younger monkeys. The results of this experiment are shown in Figure 2. Monkeys that received higher doses of stronger stem cells showed improved oral health.

How Can The Influence of Stem Cells for Longevity Influence Our Future?

Stem cells being used for longevity in the future can lead to lengthened life spans, expanded knowledge about stem cells, discoveries leading to procedures, and more ethical and less wasteful ways to help extend lifespans. Future applications include organ regeneration, disease reversal, and anti-aging therapies by using stem cells. This can lead to generations living and thriving for the life that they deserve. Stem cell breakthroughs may redefine what it means to age at all.

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A New Solution That Saves Lives Twice

Lily Bottner, '29

Horseshoe Crabs

Horseshoe crabs are one of today's most fascinating and most ancient species, having roamed the seafloors for 450 million years. They have survived extinctions, outlived the dinosaurs, and today, they are recognized for another main reason. Their blood. Aside from its intriguingly beautiful

Figure 1: A bottle of blood harvested from a live horseshoe crab at a facility in Charleston, S.C, in June of 2014

color, due to being copper-based instead of iron-based like humans (Figure 1), it has another critical use. Every vaccine and injectable medication relies on a test to ensure safety, and those tests would be nearly impossible without this animal's contribution.

Vaccine Testing and Endotoxins

In the United States, the US Food and Drug Administration (FDA) requires every drug to be approved and tested for potentially harmful bacterial contaminants (Maloney et al. 2018). Endotoxins, large molecules in the outer membranes of the cell walls of certain bacteria (Maloney et al. 2018), are released when the bacteria is killed. Endotoxins are dangerous however, as it can trigger a powerful inflammatory response in the body, causing fever or even resulting in endotoxic shock and death (GenScript 2021). However, in horseshoe crab blood, special immune cells are present that detect toxic bacteria. If those cells come into contact with endotoxins, they will proceed by forming a protective clot around it to protect the animal (Natural History Museum 2021).

A History of Testing

Before the 1970s, scientists relied on rabbits for endotoxin detection, which led to hundreds of thousands of rabbits being euthanized (The Conversation 2023). This was stopped thanks to a discovery made in the 1960s by a pathobiologist named Fredrick Bang who was studying the circulatory system of horseshoe crabs, when he noticed that some bacteria would cause the blood to coagulate. This was soon realized to be endotoxins, and thus, the Limulus Amebocyte Lysate (LAL) test was born (University of Chicago MBL 2025). In the following decade, it was implemented worldwide (University of Chicago MBL 2025) (Natural History Museum 2021). Even though the LAL test seemed to have no consequences, it was soon found out that the LAL test had major consequences for horseshoe crabs and red knot birds.

Figure 2: Horseshoe crabs being bled

Impact on Horseshoe Crabs and Red Knots

The LAL test has been exceedingly helpful for humanity by providing safe and reliable drugs, but has negatively impacted the four extant horseshoe crab populations across the globe (Humane World for Animals 2024). Every year, thousands of horseshoe crabs die from being bled, and this industry is just one piece that is contributing to horseshoe crab decline. The horseshoe crabs are drained of up to 30% of their blood (Figure 2), and after being released back into the wild, it's estimated that 30% die (Scientific American 2016). The fishing industry commonly harvests horseshoe crabs as bait for eel and whelk, and it is estimated that both the medical and baiting industries are responsible for the deaths of over half a million horseshoe crabs every year (The Conversation 2023). Because horseshoe crabs are part of their greater ecosystem, this is affecting populations that rely on them as well. In spring, when horseshoe crabs are born, millions of shorebirds migrate to feed off of their eggs (Figure 3). Red knots, a species that has an incredible migration journey from the southern tip of South America to northern Canada, make a stop along the Atlantic coast to eat horseshoe crabs so they can gain energy for the remaining trip (The Conversation 2023). Because of the decline of horseshoe crab numbers in recent decades, Red Knots were listed as threatened under the Endangered Species Act of 2015.

Figure 3: A red knot on a horseshoe crab to feed off its laid eggs

rFC and rCR

Since the 1990s, there began efforts to search for synthetic alternatives to the LAL tests. The first major breakthrough came from Singaporean researchers, who invented the first way to detect endotoxins without the use of an animal (The Conversation 2023). Using horseshoe crab DNA and recombinant DNA technology, a way to manipulate and isolate certain gene segments and generate new genes with helpful functions (NHGRI 2025), they created an alternative called recombinant Factor C (rFC) (The Conversation 2023) (Maloney et al. 2018). This success imitates the first step of the three-part LAL test. In the years following, more facilities and biomedical companies caught on and began to produce their own versions of rFC, along with recombinant Cascade Reagents (rCRs), a system of proteins that react in a specific sequence (cascade) to detect a target, which come together to simulate the entire LAL test without an ounce of horseshoe crab blood needed (The Conversation 2023). Despite these recent advancements, major biomedical testing companies still persist in using LAL testing today.

Figure 4: An endotoxin test kit with the new reagents

U.S. Pharmacopeia and The Breakthrough

U.S. Pharmacopeia (USP) is a regulatory organization that works to set safety standards for health products, and the FDA follows its instruction (NIH). It classified rFC and rCR as alternative methods to test for endotoxins in intravenous drugs. Meaning that in order for the synthetic tests to be used, it called for case-by-case validation, a process that often creates more trouble than it is worth. But just last year, in the spring of 2024, USP approved Chapter 86, titled: Bacterial Endotoxins Test Using Recombinant Reagents, a chapter on endotoxin testing, allowing biomedical testing companies the freedom to make the switch to rFC and rCR (University of Chicago MBL 2025) (Nature 2024). Because of this change, many biomedical companies are beginning the journey of switching to rFC and rCR (Figure 4). "This represents an extraordinary opportunity for the biomedical and pharmaceutical industries to significantly contribute to the conservation of horseshoe crabs and the birds that depend on them," says Maloney et al. The research shows that if the big companies convert to rFC and rCR, this will result in a 90% decrease in LAL demand, saving an estimated 100,000 horseshoe crabs in North America every year (Maloney et al. 2018) (University of Chicago MBL 2025). On top of that, switching to rFC and rCR is more reliable and sustainable, because LAL tests can lead to false positives, and are sensitive to temperature changes, pH changes, and sample impurities (BMG LABTECH 2025).

The Future

The situation today almost mirrors what happened in the 1970s, when endotoxin testing moved from rabbits to horseshoe crabs. The switch from LAL to rFC and rCR won't happen overnight, but now that it is officially approved, big strides are inevitable. On top of the animal welfare aspect, industry workers and experts have confirmed that the switch will potentially be more cost efficient (Maloney et al. 2018). Not having to fish and house the horseshoe crabs will save money, and in the coming years, synthetic testing is bound to get even better. With this step, hundreds of thousands of horseshoe crabs and red knots will be saved. This recent and necessary step is beginning the journey to helping the horseshoe crab populations recover, and while there is still a long road ahead, with further standardization of the new methods, we could help save human lives and the lives of the horseshoe crabs.

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The Discovery of HOPS-315: Our Solar System's Beginnings

Emily Cheng, '29

What is HOPS-315?

As we enter a new era of astronomical technology, scientists hope to better understand the early stages of planetary formation. The recent discovery of the young star HOPS-315 offers astronomers a rare opportunity to witness the early formation of a planetary system. This "baby star" is located 1,300 light-years away from us, and resembles our Sun billions of years ago.

In late 2023, the James Webb Space Telescope and the Atacama Large Millimeter/submillimeter Array (ALMA) identified this protostar in the Orion Molecular Cloud. After observing its development over the next few years, astronomers noticed rocky planet material in the disk around the star in July of 2025. From this, scientists deduced that the planetary system forming around HOPS-315 is nearly identical to our Solar System.

Discoveries Made

The James Webb Space Telescope and ALMA captured evidence of silicone monoxide, a gas condensing into silicate materials around the star. The materials formed can also be found as one of the most abundant components located in rocks in the Earth's crust. In addition, both a disk of gaseous and crystalline silica formed (Figure 1), confirming that the planet was in its earliest stage of formation. Logan Francis, an astronomer at Leiden University, states, "We're really seeing these minerals at the same location in this extrasolar system as where we see them in asteroids in the Solar System." The asteroids in our Solar System had previously been our only indication of the involvement of silicon in planet building previously. However, with the new evidence from HOPS, astronomers are able to confidently conclude that the silicon combines into small pebbles, creating the fundamental building blocks for rocky planets.

In addition, ALMA's detection of silicon monoxide, bipolar gas jets (Figure 2) allows astronomers to understand the direction of mineral flow in an undeveloped star's protoplanetary disks. While they confirm that HOPS-315 is very young, the gas jets also reveal angular momentum and magnetic field strength in protostars. Since the bipolar gas jets shoot out in opposite directions, they align closely with the star's rotational field. The jets reduce the amount of angular momentum, allowing more silica into the orbit of the star, and thus shaping its planetary disk in the process.

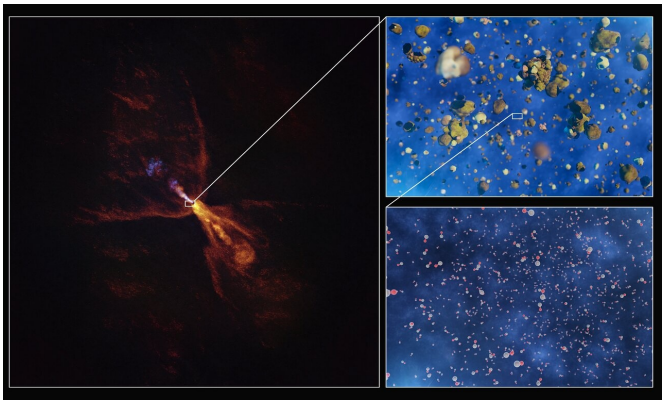

Figure 1: In the leftmost image, the orange shows gaseous silicon blowing away from HOPS-315. The images on the right indicates crystalline silica clumping together in its protoplanetary disk (Alma Observatory)

Significance of HOPS-315

According to Merel van 't Hoff, a co-professor at Purdue University, the planetary system involving HOPS-315 is what our Solar System looked like when it was beginning to form. HOPS will be extremely beneficial in helping astronomers

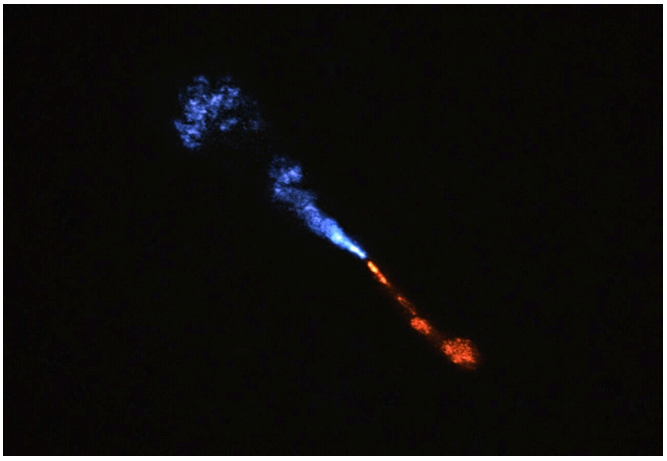

Figure 2: The bipolar, silicon monoxide jets blasting away from the star. The blue jet is blowing towards the Earth, and the orange is blowing away from us (European Southern Observatory).

find how early chemical compounds influence the formation of planets.

Due to HOPS' striking similarities with our own Sun, astronomers expect that observing the process of planetary formation around HOPS will allow us to refine the theories of how the Solar System and rocky planets are formed.

Additionally, seeing the process of silica changing into gaseous and crystalline forms will allow scientists to grasp the concept of compounds clumping and condensing into larger bodies.

Ultimately, the discovery of HOPS-315 will be a fundamental resource to enable astronomers to fully understand the origins of our Solar System. As further research is done about it, scientists hope to reveal and confirm more theories about the formation of rocky planets and the significance of the compounds created prior to their full development.

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For the Third Time Ever an Interstellar Object is Spotted

Luke Cooper '27

On July 1st, 2025 3I Atlas was discovered for the first time by the Asteroid Terrestrial-impact Last Alert System (ATLAS) telescope in Chile. According to NASA, the orbital path is hyperbolic meaning that, "it is simply passing through our solar system and will continue its journey into interstellar space, never to be seen again."(NASA) In other words, 3I/ATLAS is an interstellar object. The third ever observed entering our solar system.

About 3I Atlas

The previous two interstellar objects were Oumuamua which was spotted in 2017, and Comet Borisov which was spotted in 2019. Both Oumanmua and Comet Borisov were big deals but what makes 3I Atlas even more special is its size and its origin. Omunamuam, the first interstellar object, was observed to be a little more than 200 meters. Comet Borisov, on the other hand, was observed to be 1km long. Early observations from the Hubble Space Telescope (HST) estimated that the size of 3I/ATLAS's is up to 3.5 miles (5.6 km) in diameter. This makes it more than five times larger than the previous two interstellar objects. Also, 3I Atlas is moving much faster than Oumanmua and Comet Borisov. According to NASA visualization it is "racing through our solar system at 130,000 mph".

Astronomers using the Very Large Telescope (VLT) in Chile found bright nickel vapour in the gas surrounding the comet at a distance ~3.88 au(1 au equals the distance between Earth and Sun) where temperatures are normally too cold for metals to vaporise. They write, "In a surprising discovery… we detected glowing nickel vapor in the gas surrounding the incoming interstellar comet 3I/ATLAS at an extraordinary distance from the sun, where temperatures remain far too cold for metals to normally vaporize." Furthermore, the cloud of gas/dust around the nucleus (the coma) is unusually rich in carbon dioxide (CO_2). A James Webb Space Telescope (JWST) study found a CO_2/H_2O mixing ratio of about 8:1, "among the highest ever observed in a comet"(NASA).

The unusually high amount of CO_2, early nickel release, and very fast speed indicates that 3I Atlas comes from the thick disk in the Milky Way(Figure 2). The thick disk of the Milky Way is an older layer of stars that surrounds the galaxy's thin disk. Its stars are typically 8–12 billion years old, made up of light elements(Hydrogen and Helium), and orbit the galaxy at a slight tilt, giving the disk its thicker appearance.

Astronomers think the thick disk may have formed early in the Milky Way's history making it a record of the galaxy's ancient past. This means that by studying the 3I Atlas we can learn more about the thick disk, therefore allowing us to learn more about the creation of our galaxy.

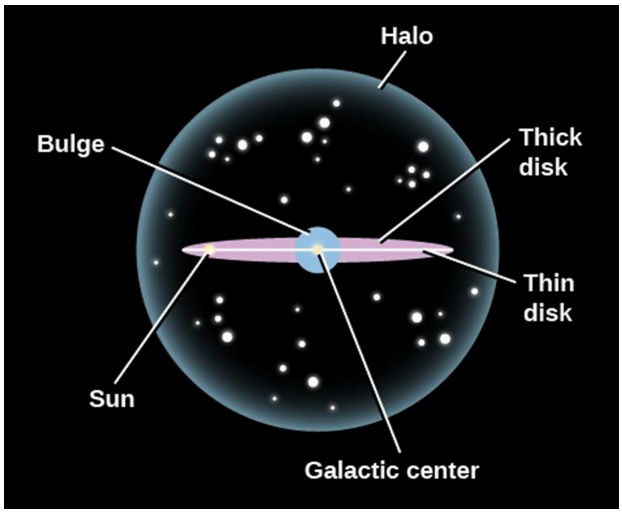

Figure 1: Image showing different parts of the Galaxy including the thick disk(Cal Poly)

What NASA is Doing

3I/ATLAS is making a one-time pass through our solar system. According to NASA's blog, it will reach its closest point to the Sun on October 30th 2025 at a distance of about 1.4 au (distance from Earth to Sun). Because of this, it poses no threat to Earth. NASA confirmed that it will "remain at a distance of at least 1.6au" from Earth.

Because 3I/ATLAS is fast-moving and will not get close to Earth, scientists cannot launch a dedicated mission in time. Instead, they are adapting existing spacecraft and observatories. For example, ESA's orbiters at Mars the ExoMars Trace Gas Orbiter (TGO) and Mars Express are being used to observe the comet when it passes by Mars in October 2025. The spacecraft had "a unique vantage point" for observation. (Sky at Night Magazine) Similarly, researchers pointed out that NASA's Jupiter-bound probe could potentially detect charged particles from 3I/ATLAS's tail, describing the opportunity as "literally a once-in-a-lifetime opportunity." (NASA)

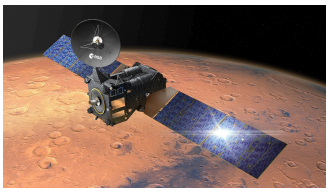

Figure 2: Image of the ExoMars Trace Gas Orbiter. A satellite orbiting Mars that's goal is to find life on Mars. However, it is now being repurposed to study 3I Atlas. (European Space Agency)

Current Status

As of late 2025, 3I/ATLAS has passed the sun and is moving outward, but it is still under careful observation. Its evolving tail, coma, brightness and trajectory are being monitored. According to a Newsweek live update: "The comet… has displayed erratic changes in speed, brightness, and trajectory, prompting global scientific concern" (Newsweekly). The shift in color and activity suggests the comet is entering a new phase, the active phase. This is happening because 3I Atlas is heating up from the sun causing more gasses to be released. By analyzing the gasses released scientist will be able to figure out the exact composition of 3I Atlas as well as the layering inside 3I Atlas. In terms of visibility, the object is faint(Figure 4). Early reports stated that while it may be detectable with reasonably sized amateur telescopes in late 2025 and early 2026, it will not be visible to the naked eye. Because of its hyperbolic trajectory, 3I/ATLAS will leave the solar system and never return. That means all observations must be taken now, "before it disappears back into the cosmic dark." (NASA)

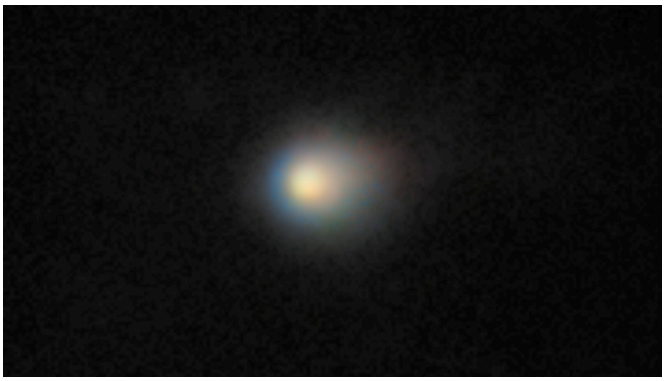

Figure 3: This Image shows 3I Atlas from the Gemini North Telescope in Hawaii. (Space.com, Mathewson)

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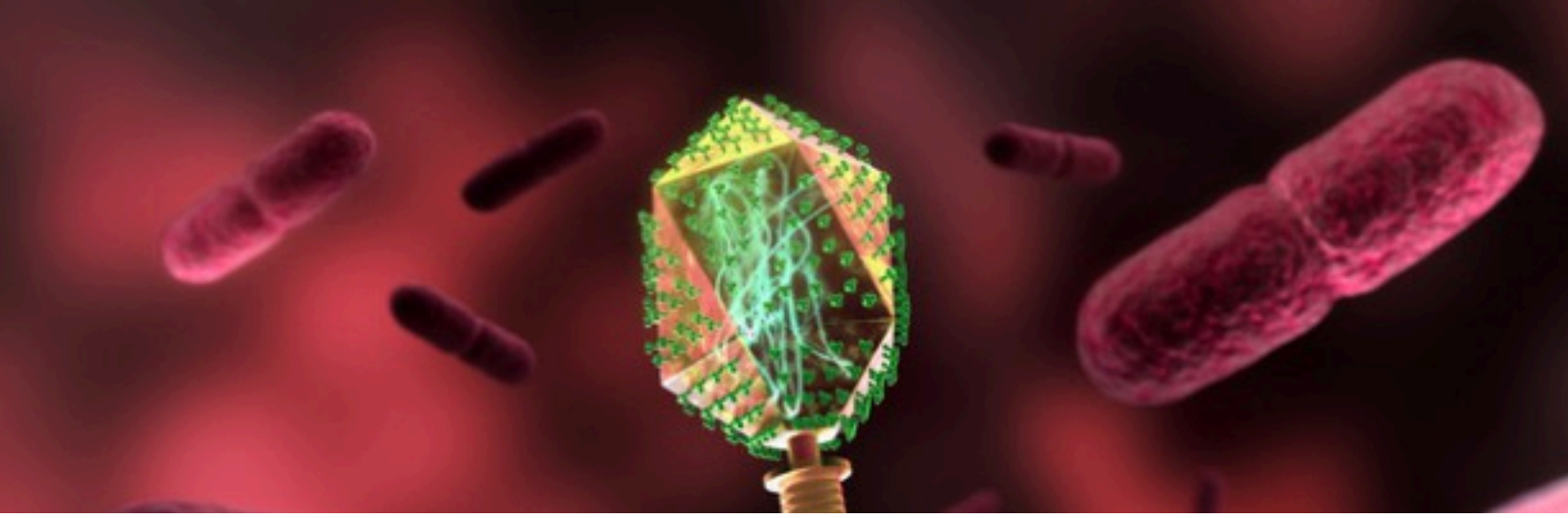

Blueprints of Life: AI's First Designed Viruses

Alexandra Dalamagas, '28

AI-Designed Life

For the first time in history, artificial intelligence (AI) has developed living viruses—demonstrating that AI can now create life from scratch. As of September 17th, scientists from Stanford University and the Arc Institute have created two AI models, Evo-1 and Evo-2, capable of generating blueprints for viruses (McCarty, 2025). These viruses, called bacteriophages, halted the growth of E. coli strains, laying the foundation for future phage therapies to target and treat microbial infections. Modeled off the bacteriophage ΦX174, 16 of the 300 AI-generated phages successfully infected E. coli, demonstrating their ability to kill bacterial strains that had already developed resistance (Garcia de Jesus, 2025).

AI Models Create the First Working Virus Genomes

Using two language genome models, scientists trained Evo-1 and Evo-2 to pair billions of genetic nucleotides (i.e., A, C, G, T) to design blueprints for more than 300 bacteriophages, 16 of which functioned successfully. The bacteriophages were modeled after ΦX174 (Figure 1), which was first discovered in the sewers of Paris in 1935. Because these phages infect only bacteria, they pose no risk of causing disease or an outbreak.

ΦX174 is easy to work with because it contains a small, simple genome and infects a nonpathogenic strain of E. coli. The AI-generated bacteriophages infected and killed E. coli in petri dishes, with some strains killing the bacteria faster than ΦX174 itself. Notably, Evo-Φ69, a variant, outperformed ΦX174, growing to 65 times its starting amount (McCarty, 2025). Additionally, the researchers did not train the AI to design human pathogens; instead, their goal was to test whether AI could generate living biological entities.

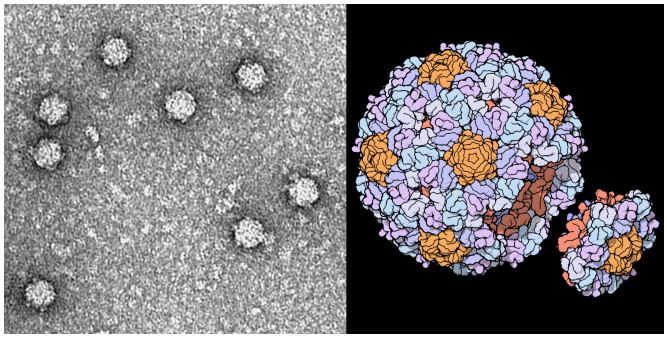

Figure 1: A micrograph of ΦX174 phages (left) and the 3D structure of a single ΦX174 phage (right) [Image]. (McCarty, 2025).

How Did AI Learn to Build Entire Genomes?

Prior to the study, the AI models were trained on 9.3 trillion nucleotides of DNA from 128,000 organisms to learn how to construct functional viral genomes (King et al., 2025). By learning patterns across all domains of life, the models could even generate the missing stretches of the ΦX174 genome. AI-generated phages infect E. coli through a series of steps. The proteins on the outside of each ΦX174 phage latch onto E. coli cells, allowing the phage to inject DNA into the host. Once inside, the bacterium uses its own protein machinery to replicate the phage's DNA. The new phages then burst through the host's cell membrane, killing it, and go on to collide with and infect additional E. coli cells (McCarty, 2025). Overall, the AI-generated phages showed higher fitness than natural ones. They not only overcame E. coli resistance but also

displayed rapid evolution.

Medical Uses & Ethical Concerns

AI-generated phages may have major potential in medicine, especially for phage therapy to target difficult microbial infections. Because these phages do not infect people, they offer a safe and practical model for lab work. AI-designed viruses or microbes could also be engineered to accelerate processes such as antibiotic production or enhance the ability of microbes (Figure 2) to degrade plastics (Garcia de Jesus, 2025).

However, scientists have raised concerns over the ethicality of AI in generating genomes. For example, many have speculated on the risk of programming Evo-models to design human viruses, such as the HIV genome (McCarty, 2025). This raises the need for strict control, oversight, and extensive testing before using these technologies. Still, the gap between digital designs and real biological assembly limits AI's ability to produce large or complex organisms.

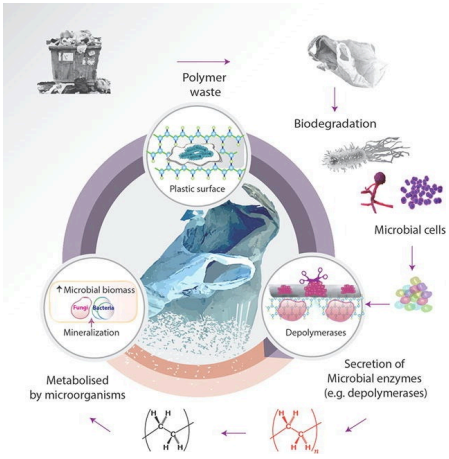

Figure 2: Process of microbial degradation, which could be sped up with the introduction of AI-designed microbes [Diagram]. (Zeenat, Elahi, Bukhari, Shamim, & Rehman, 2021).

AI-Made Viruses in the Future

The success of AI-generated phages marks the beginning of a new technological era. Scientists strive to improve these genome-designing models further. For instance, they aim to develop a "training feedback loop" to enable AI models to produce phages with predetermined characteristics, advancing current technology (King et al., 2025). However, the cost and production of bacterial genomes remain a concern for biotechnology companies. Small phage genomes in a laboratory are inexpensive to produce, but constructing hundreds of large bacterial genomes would put most biotechnology companies out of business. Nonetheless, the breakthroughs achieved with AI-designed phages reveal the potential of artificial intelligence to transform biotechnology, medicine, and our understanding of life itself.

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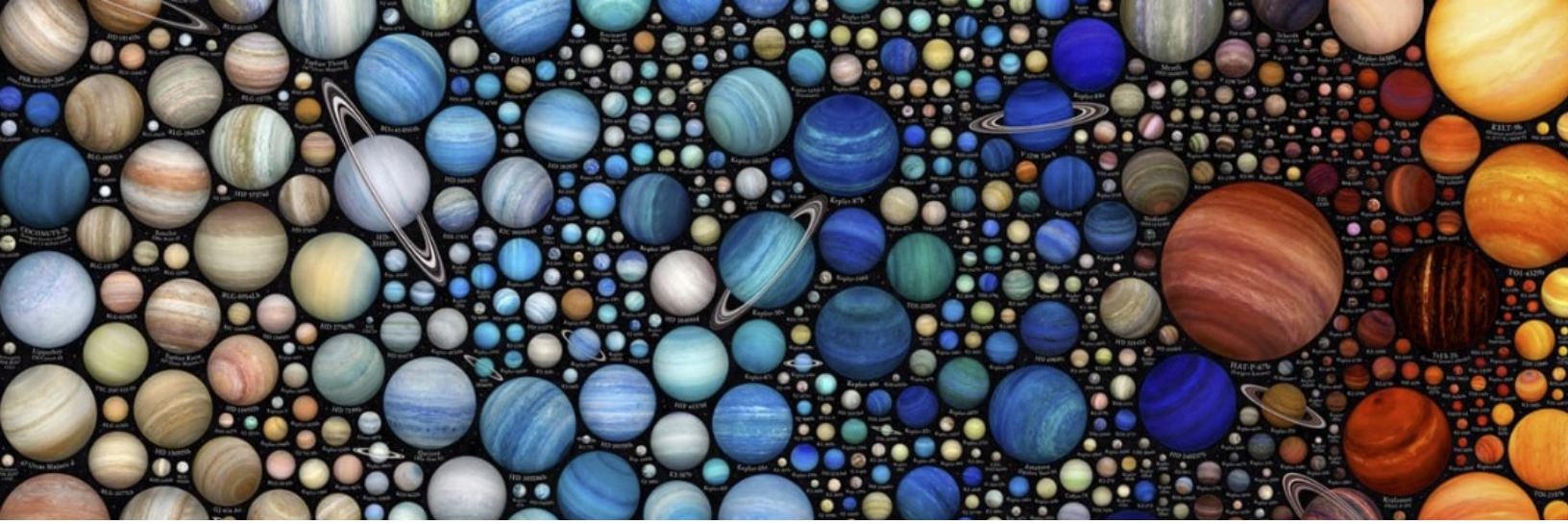

Ground-Based Light Curve Observations: Investigating Exoplanet Candidate TOI 5352.01

Serena Diamond '28

Author's Note

This study was conducted by Serena Diamond '28 in collaboration with Professor Peter Plavchan and the exoplanet research group at George Mason University (GMU). Observations and analysis presented here are part of a professional effort to verify and characterize planet candidates identified by NASA's TESS mission.

Background

Exoplanets, planets that orbit stars beyond our solar system, have quickly become one of the most captivating topics in modern astronomy. To date, astronomers have confirmed almost 6,000 exoplanets, which has given them a remarkable knowledge of planetary systems throughout the galaxy (NASA, 2025). One of the most effective detection techniques is the transit method, which measures periodic dips in a star's brightness as an orbiting planet passes in front of it. NASA's Transiting Exoplanet Survey Satellite (TESS) has discovered thousands of exoplanet candidates using the transit method. However, not all transit-like signals indicate actual planets; some may result from binary star systems, instrumental artifacts, or other false positives. To discern which signals actually indicate planets, astronomers depend on follow-up observations taken from Earth, called ground-based observations. These follow-ups are essential in confirming that scientists are truly observing a transit.

Methods

This study focused on TESS Object of Interest (TOI) 5352.01, an exoplanet candidate theorized to have an estimated radius 8.7 times that of Earth's and an orbital period of 2.04 days (ExoFOP-TESS, 2025). Observations were conducted on December 4th, 2023 using a 0.8-meter telescope with a red filter and 90-second exposures. The collected images were processed using AstroImageJ (AIJ), a software for astronomical image processing (Collins et al., 2017).

All images underwent dark and flat-field calibration to correct for instrumental noise and uneven illumination. They were then aligned and analyzed using differential photometry, which compares the brightness of the target star to nearby stars that stay constant, helping spot small changes in light. Each image was matched to exact sky coordinates (plate-solving), and the data was used to graph the light curve, see Figure 1, showing how the star's brightness changes over time (Diamond & Plavchan, 2025).

If this candidate were an exoplanet, a dip in brightness would have been expected during the predicted transit window. However, analysis of the light curve revealed no statistically

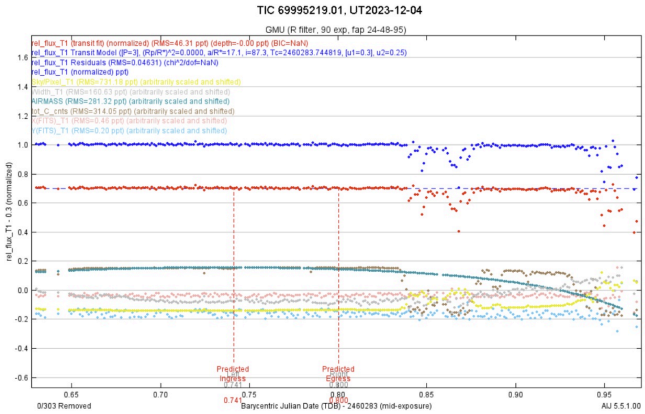

Figure 1: Light curve of TOI 5352.01 from GMU's ground-based telescope observations demonstrating no visible transit. [Graph] (Diamond & Plavchan, 2025).

significant decrease in the star's flux excluding a time after the predicted transit when the entire frame's flux decreased, most likely due to cloud cover.

A Nearby Eclipsing Binary (NEB) check was conducted to ensure that the observed light curve was not interfered with, see Figure 2. No stars flagged in the NEB analysis were close enough to the target star to interfere with its light curve. These results indicate that TOI 5352.01 is a false positive and that the dimming detected by TESS was not caused by a planetary transit.

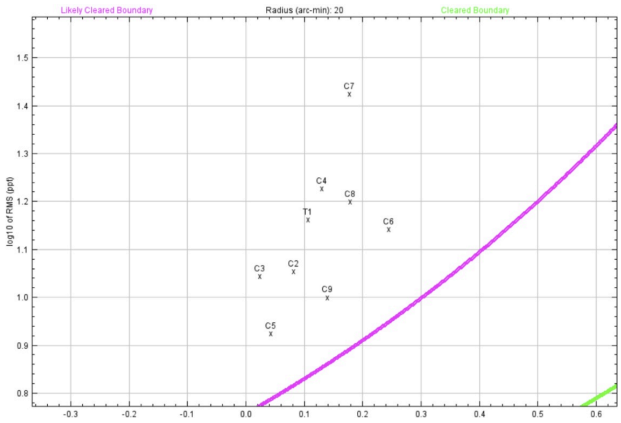

Figure 2: NEB analysis from GMU observation of TOI 5352.01. All candidates were near the likely cleared boundary and were far from the transit star in the observed frame. [Graph] (Diamond & Plavchan, 2025).

Significance

Exoplanet research is expanding scientists' understanding of planetary systems and the potential for life beyond Earth. In this study, no exoplanet was confirmed, demonstrating the critical role of ground-based verification in refining space-based survey data. Follow-up observations can contradict initial findings and improve the reliability of the TESS candidate catalog, helping astronomers rule out false detections and better characterize real planetary signals.

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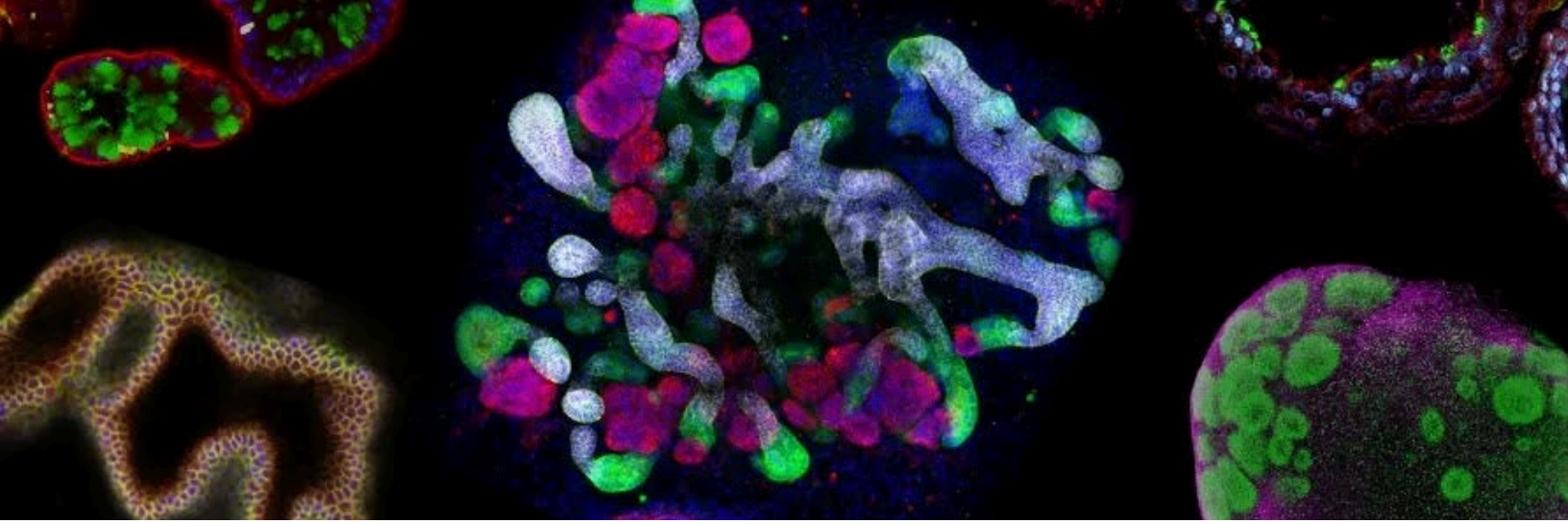

Mini Organs: How Organoids are Changing Biomedical Research

Natalia Garment (‘26)

The Role of Organoids in Modern Biomedical Research

Organoids are small, three-dimensional structures grown from stem cells that resemble mini versions of human organs. They can organize themselves into complex tissues that act and respond like real organs. Organoids are derived from cells that have stem potential, meaning they can divide indefinitely and produce different types of specialized cells, such as liver, intestine, and brain cells. Scientists add specific growth factors, signaling molecules, and nutrients to the culture medium to direct differentiation (Harkin et al. 2024). Once placed in the right environment, these cells self-organize into a three-dimensional mass and follow their genetic instructions to form structures that resemble organs in natural development (Figure 1). Because organoids behave so similarly to human tissue, they have become an important tool for studying how diseases develop and how new drug treatments might work.

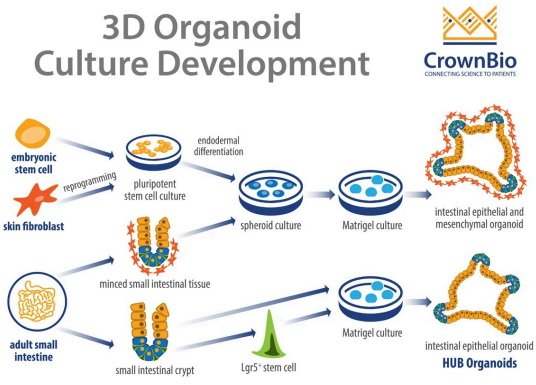

Figure 1: Development and differentiation of intestinal organoids

Disease Modeling and Basic Research

One of the main uses of organoids is to model how organs develop and diseases progress. Traditional cell cultures grow in two-dimensional layers, which do not accurately represent how cells behave inside the human body (Yang et al. 2017). Animal models also have limitations because human diseases do not always act the same across species. Organoids solve these problems by recreating human tissue in the three-dimensional form.

Researchers use organoids to study human development and organ formation (organogenesis). For example, organoids grown from stem cells can show how the brain, bone, heart, or liver forms in early development (Barbuzano 2017). This helps scientists better understand developmental disorders and genetic defects.

Organoids are also powerful models for complex diseases. For instance, gastrointestinal organoids help scientists study inflammatory bowel disease and identify subtypes that predict how patients will respond to treatment (Buckles 2024). Liver and pancreatic organoids are used to explore genetic disorders such as Wilson’s disease, cystic fibrosis, and diabetes-related conditions. Brain organoids allow researchers to investigate disorders like autism, schizophrenia, and microcephaly, which are hard to study using animals (Barbuzano 2017).

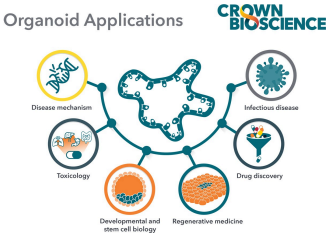

Figure 2: Illustration of the key applications for organoids

Infectious disease research also benefits greatly from organoid technology. Because some viruses and bacteria only infect human cells, organoids can provide more accurate models than animals. Scientists have used organoids to study infections caused by COVID-19, hepatitus, Zika, and Helicobacter pylori (Yang et al. 2023)

Drug Discovery, Screening, Toxicology

Organoids are revolutionizing the drug development process. Because they closely resemble real human organs, organoids let scientists test new drugs on "mini-organs" before clinical trials. This allows researchers to screen drugs for effectiveness and safety early on, saving time and money while reducing animal testing.

For example, organoids have been used to test drugs that might protect brain cells from Zika virus infection or correct problems caused by cystic fibrosis (Barbuzano 2017). They also help predict side effects by modeling how different organs respond to chemicals. Liver organoids are used to test for drug-induced liver injury, while kidney organoids assess potential nephrotoxicity (Yang et al. 2023).

Regenerative Medicine and Tissue Repair

In the long term, organoids may help replace or repair damaged tissues, reducing the need for donor organs. Scientists have already transplanted liver organoids into mice, improving liver function and survival rates (Yang et al. 2023). Similarly, pancreatic organoids have shown potential to restore insulin production in diabetic models, offering hope for future diabetes treatments (Barbuzano 2017).

Other examples include using bile duct organoids to repair damaged ducts and cerebral organoids to help restore brain tissue after stroke or injury. As these technologies advance, organoids may serve as building blocks for tissue reconstruction and even full organ replacement.

Conclusion

Organoids represent a major breakthrough in biomedical research. They bridge the gap between simple cell cultures and complex human biology, offering new avenues to study disease, drugs, and regenerative medicine, showing the potential to reduce the need for organ donors.

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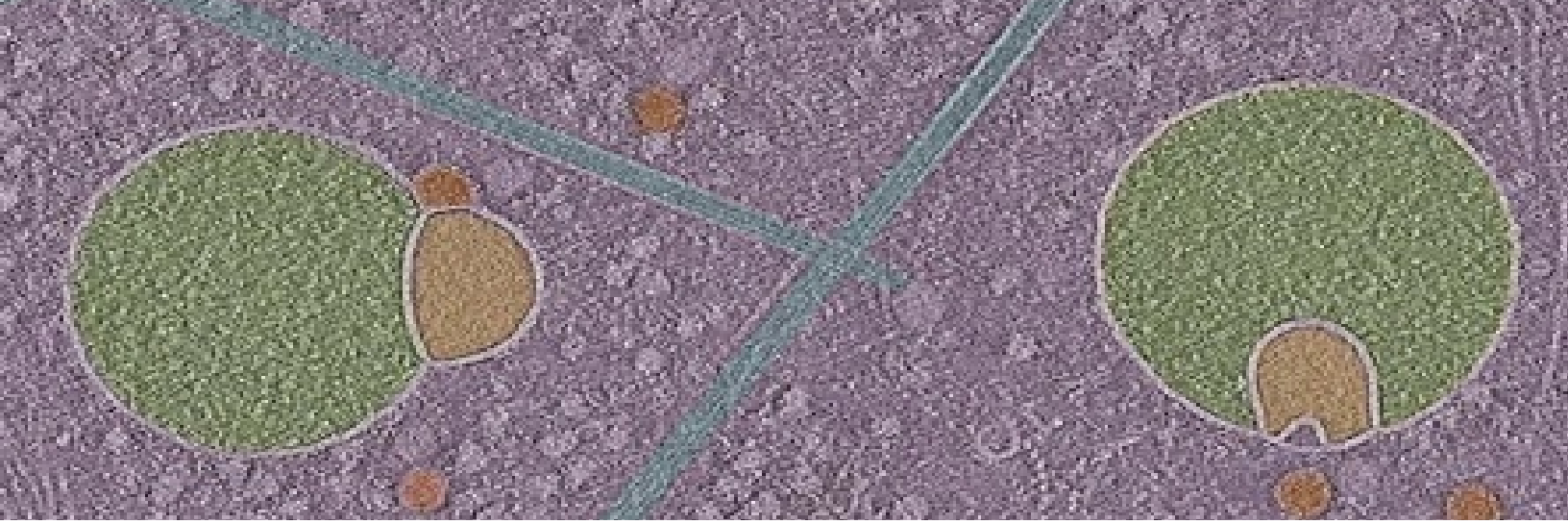

The Discovery of Hemifusomes

Adrianna Gavriloiu, '29

Introduction: Overview of Hemifusome Discovery

This past June, researchers from the University of Virginia and the National Institutes of Health worked at UVA's Molecular Electron Microscopy Core, an ultramodern cryo-electron microscopy (CEM) facility, co-led by Seham Ebrahim, PhD, as assistant professor in the Department of Molecular Physiology and Biological Physics at UVA, and Bechara Kachar, MD, the chief of the Laboratory of Cell Structure and Dynamics at the NIH's National Institute on Deafness and Other Communication Disorders, discovered a new organelle: hemifusomes (Aminuddin, 2025). These cellular structures and their interacting proteolipid nanodroplets are an essential step in the creation of multivesicular bodies (MVBs) in a lipid remodeling pathway separate from the previously known ESCRT process. The produced MVBs are essential sorting and recycling centers within cells. Therefore, they are critical for immune function and communication between cells; however, they can cause neurological diseases when functioning incorrectly. Fortunately, this enables the discovery of hemifusomes to act as a key step in better understanding Alzheimer's, viral infections, certain cancers, and other illnesses (Aminuddin, 2025). To begin breaking down these complex processes and functions, understanding the microscopy techniques used in this discovery is essential.

Technique: Cryo-Electron Tomography

Hemifusomes were first visualized using cryo-electron tomography, a specific type of cryogenic electron microscopy (cryo-EM). In cryo-EM, molecular samples are first purified to eliminate any contamination, then dissolved into an aqueous solution, a mixture where water is the solvent, to allow free movement. A tiny sample of the solution is spread across a fine metal mesh, which is then shot down in liquid ethane by a robotic arm. Liquid ethane is used because of its incredibly low freezing point at -188°C, allowing it to rapidly drop the temperature of the solution. Because the temperature change happens so quickly, the water does not form ice crystals that would destroy the sample, but instead becomes vitrified into a glassy structure that keeps the molecules intact. The frozen sample is then shot with an electron beam in a vacuum chamber because of its tiny wavelength, allowing for detailed images on an incredibly small scale. The process in which a 3D rendering is then created is specific to cryo-electron tomography. A singular molecule or structure being photographed from multiple angles is used to compile many 2D images used for generating the final product (Chui, n.d.). This technique was essential to visualize a structure as small as a hemifusome, which would have been impossible using only conventional microscopy. To further realize the significance this discovery had on knowledge regarding MVBs, understanding the ESCRT pathway, previously believed to be the primary method of MVB formation, is crucial for context.

Context: ESCRT Pathway

Endosomal sorting complexes required for transport (ESCRT) involves protein scaffolding. Protein scaffolding is the use of scaffold proteins as hubs, bringing together and coordinating smaller proteins in order to increase efficiency and send out biochemical signals (DiRusso et al., 2022). ESCRT, specifically, is divided into five complexes (parts): ESCRT-0, ESCRT-I, ESCRT-II, ESCRT-III, and the Vps4

complex, each of which are proteins. To begin, ESCRT-0 collects ubiquitinated cargo, meaning it has been bonded to ubiquitin, a small regulatory protein, marking it for degradation. The cargo then bonds to the endosomal membrane, the boundary between the cell's cytosol and the lumen of an endosome. Cytosol is the fluid inside of cells, whereas the lumen is the interior space of a membrane bound vesicle. These vesicles are liquid-filled sacs that protect cellular components during transport across the cytoplasm, acting like delivery trucks. In the context of MVBs, these endosomes are also the home of smaller intraluminal vesicles forming inside of them, hence the name multivesicular bodies. ESCRT-I then binds to both the ESCRT-0 and the cargo, causing the membrane to bend inward, beginning a bud that will become an intraluminal vesicle. ESCRT-II further bends the membrane, deepening the bud and activating ESCRT-III. ESCRT-III proteins floating in the cytoplasm assemble into helical chains in the neck of the bud, finally cutting the vesicle off and releasing it into the endosome. The Vps4 complex, an enzyme powered by ATP energy, then deconstructs the ESCRT-III proteins left at the budding site, sending them back into the cytoplasm for reuse (Schmidt & Teis, 2012).This repeating process causes multiple smaller vesicles to form inside of an endosome, creating a multivesicular body. The diagram below illustrates the essential steps of ESCRT. However, the discovery of hemifusomes has shown that this complex process is not the only method of creating an MVB.

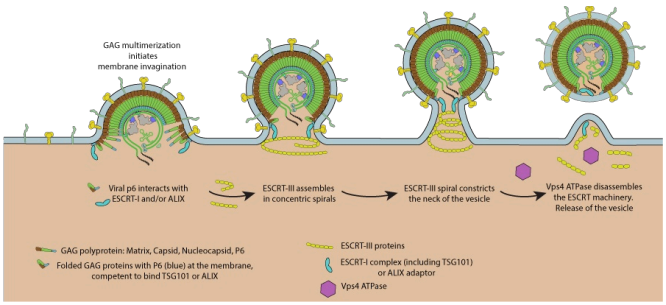

Figure 2: ESCRT separating a vesicle from the budding site [Illustration]. (Viral Budding, n.d.)

New Understanding: Lipid Remodeling

MVBs can additionally be formed through lipid remodeling, a process in which vesicles are created without the ESCRT pathway. Endosomal membranes are made of phospholipid bilayers: two phospholipids, a subcategory of lipids, set back-to-back to create a boundary between the liquids in different vesicles and the cytosol due to their hydrophobic nature (Chernomordik, 2008). To begin, a class of enzymes known as lipid kinases alter the chemical composition, and therefore properties, of the endosomal membrane in one area through phosphorylation: the addition of phosphate, typically from an ATP molecule. A category of cone-shaped lipids then naturally cause inward curvature in that area. This creates a bud, which is then cut off to form a vesicle, which can happen with or without the help of ESCRT-III proteins (Kozlovsky et al., 2002). However, before this final stage, hemifusomes are formed. Hemifusomes are partially-fused intermediate structures, where the outer layer of the phospholipid bilayer is still merged together, but the inner compartment of the vesicle has split off to line the lumen of the new vesicle. The proteolipid nanodroplets, which are lipid and protein rich particles, bud off from the high-curvature area to release physical stress on the twisting and tightening membrane (Aminuddin, 2025). These structures provide proof of the use of lipid remodeling in the formation of MVBs, and deepen understanding of these critical endosomes.

Future: Importance of Hemifusomes and MVBs

Through both of these methods of formation, multivesicular bodies are able to accomplish incredible feats for the cell, and keep the degradation, storage, and recycling of cellular materials in check. The vesicles inside of them, intraluminal vesicles, contain cellular cargo that can be transported to lysosomes, organelles that break down used molecules to be recycled into new materials, or to exosomes, extracellular organelles that exit the cell to transfer intercellular information (Turegano-Lopez et al., 2019). The critical job of MVBs in the body consequently enables their newly deepened comprehension to advance understanding of fatal diseases, propelling medicine and scientific knowledge forward.

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Compound 12b: How the Use of Artificial Intelligence is Transforming Cancer Drug Design

Jade Lan, '29

On February 26th, 2025, researchers at Insilico Medicine (a clinical-stage biotechnology company driven by the use of artificial intelligence) announced the creation of a covalent dual inhibitor called compound 12b. Using their own PandaOmics and Chemistry42 artificial intelligence (AI) platforms, compound 12b is designed to block DNA repair pathways that cancer cells rely on to survive. By targeting cyclin-dependent kinases 12 and 13 (CDK12/13), the compound prevents tumor cells from repairing treatment-induced damage, making them more vulnerable to existing therapies such as chemotherapy. This AI-driven breakthrough not only offers a promising new way to weaken cancer cells but also demonstrates how artificial intelligence can speed up and refine the process of developing next-generation cancer treatments.

What is CDK12/13?

Cyclin-dependent kinases (CDKs) are enzymes that must bind to a cyclin protein before they can phosphorylate target proteins. By adding phosphate groups, CDKs advance cell division and support DNA repair, making them essential for healthy cellular function. However, the overexpression of CDKs 12 and 13 has been shown to promote malignant tumor development by activating signaling pathways that support the survival of cancer cells (Dai, 2025). For instance, CDK12 promotes the migration of HER2-positive cancer cells by influencing pathways like P13/AKT, which regulate cell proliferation and allow cancer cells to spread more aggressively.

What is Compound 12b?

Compound 12b is a small-molecule drug designed to simultaneously inhibit CDKs 12 and 13 by forming covalent bonds at their active sites (Insilico Medicine, 2025). This blocks DNA repair in cancer cells, which they rely on to survive errors caused by rapid cell division and treatment-induced damage. As a result, the cancer cells accumulate DNA breaks and stop proliferating. Lab studies have shown that compound 12b has high nanomolar potency and, in combination with existing treatments, can suppress tumor growth in over 6 types of cancer (including difficult-to-treat cancers like triple-negative breast cancer and acute myeloid leukemia). Researchers also found that compound 12b was highly selective, targeting CDK12/13 without disrupting other kinases. While healthy cells also rely on CDK12/13 to maintain genomic stability, they aren't impacted as much by the inhibitor because they divide less frequently and experience less DNA stress than cancer cells. The selectivity of compound 12b has helped minimize off-target effects and toxicity, which have been major challenges in earlier CDK inhibitors.

How Was AI Used in the Creation Process?

Insilico Medicine used two of its own AI platforms — PandaOmics and Chemistry42—to accelerate the creation of compound 12b. PandaOmics first identified CDK12 and CDK13 as valuable drug targets by analyzing biological datasets and ranking genes based on their relevance to tumor development and druggability. The platform used data on gene expression, methylation, proteomics (protein levels), and mutations, then evaluated targets using both basic bioinformatic methods and advanced machine learning tools, including a heterogeneous graph walk algorithm that searched for genes closely related to the disease (Kamya, 2024). Once CDKs 12 and 13 were selected, Chemistry42, a generative

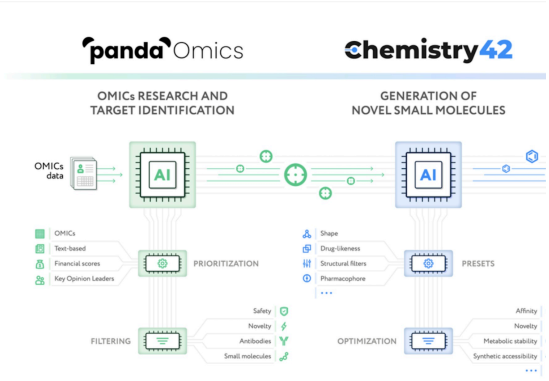

Figure 1: Insilico's AI pipeline for drug discovery. (Insilico Medicine, 2022).

chemistry platform, designed new small molecules that could bind specifically to these kinases (Figure 1). The platform used generative AI to propose possible structures, then filtered through them based on predicted binding ability, stability, selectivity, and safety. It explored chemical variations through hit identification, R-group exploration, and scaffold hopping, then optimized key ADMET properties to improve the chances that the molecules would work in the human body.

What Does This Mean for the Future of Medicine?

The development of compound 12b showcases how AI can transform the drug creation process. By integrating AI into the earliest research stages, scientists can predict molecular structures, simulate biological interactions, and rapidly evaluate huge numbers of potential drug candidates, compressing work that once took several years into a little over one year, and speeding up early discovery by an estimated 40 to 50 percent. This helps researchers find new therapeutic targets and design safer, more effective treatments, shortening the long and expensive path to clinical testing (Figure 2). Insilico isn't the only company moving in this direction; other groups, such as Exscientia, are also developing AI-designed drugs for conditions ranging from OCD to rare genetic and fibrotic diseases. In oncology specifically, AI-guided design opens the door to more personalized therapies tailored to the unique genetic features of each patient's tumor. Although compound 12b is still in the preclinical phase and has not yet been tested in humans, its success so far suggests that similar AI-designed drugs could reach clinical trials in the near future. If successful, compound 12b could become a model for how advanced AI systems and human expertise can work together to tackle complex diseases—offering new hope for patients and reshaping the future of modern medicine.

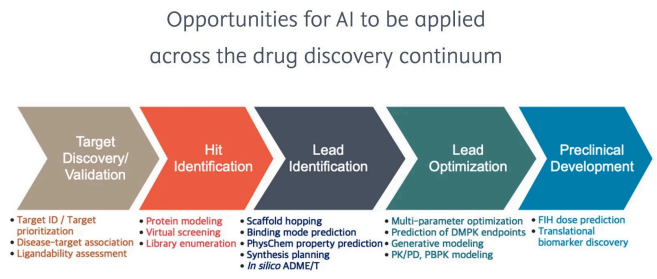

Figure 2: Opportunities for the use of AI in drug development. (Drug Discovery Today, 2022)

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The Cutting Edge of Agriculture: How CRISPR Could Redefine Food and Farming

Lyla McEntee, '27

Understanding Gene Editing Techniques

In an era marked by climate change, rapid population growth, and widespread food insecurity, the need for innovative agricultural solutions is more vital than ever before. One of the most promising advancements in this domain is CRISPR, or Clustered Regularly Interspaced Short Palindromic Repeats. While CRISPR can modify the DNA of plants and thus food, it is also used in many other contexts, such as for studying diseases, developing treatments for diseases, and modifying animals to become disease-resistant. CRISPR, the sequence found within bacteria, is designed to help bacteria fight viruses and was originally discovered in 1987. By 2012, scientists had adapted the system they found in the bacteria for precise DNA editing. This groundbreaking gene-editing tool enabled alterations to an organism's DNA (Figure 1). This is what distinguishes CRISPR from traditionally genetically modified organisms (GMOs) within the realm of food. A traditional GMO is made through the transfer of the desired or "superior" gene from one species to another species, allowing the species with the transferred gene to possess the new trait (Stitzer). GMOs are also different from genetically modified (GM) crops, which are defined as crops that have been genetically edited using genetic engineering techniques (Figure 2). This means that GM can include food altered by CRISPR if it is used to introduce foreign DNA to a species. It is important to understand that while a GM crop can be classified as a GMO, traditionally modified GMOs differ significantly from those edited using CRISPR technology, limited within a singular species. CRISPR can modify genes within the same species without introducing foreign DNA, differentiating it from a GMO and paving the way for safer and more effective crop modification (Figure 2).

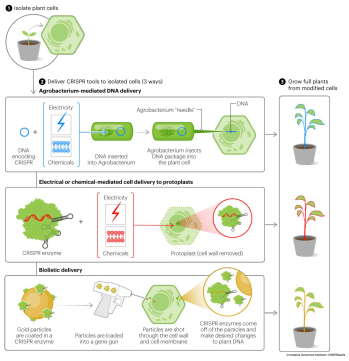

Figure 1: A guide to the three different ways to modify foods using CRISPR technology (Innovative Genomics Institute).

CRISPR is revolutionizing food enhancement by improving nutritional value, shelf life, disease resistance, and environmental resilience, with applications that include increasing crop yields in response to climate stress and tailoring plants to meet specific dietary needs. The use of CRISPR in food modification provides a groundbreaking approach to enhancing the nutritional value, safety, and resilience of food crops to address global challenges such as food insecurity and climate change.

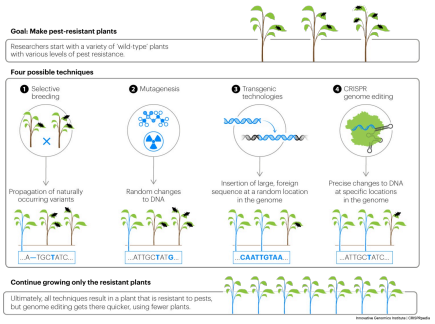

Figure 2: A differentiation between genetic editing techniques, including selective breeding, mutagenesis, transgenic technologies, and CRISPR genome editing (Innovative Genomics Institute).

The Impact of CRISPR on Crop Modification

CRISPR technology presents a revolutionary solution for

addressing global challenges such as hunger, poverty, and climate change by enhancing food security, nutritional value, and agricultural efficiency. In 2024, the Alliance For Science described how "[CRISPR] could unlock new opportunities for low- and middle-income countries to address hunger, combat poverty and inequality, and even cut carbon emissions" (Devex 2). GMOs themselves are great for fighting the growing problems of climate change, population growth, and disease likelihood. While they are the best option readily available, CRISPR seems to be even more efficient, effective, and safe. This is furthered by Ahmad, Aftab, et al. in a peer-reviewed article about the difference between CRISPR and GMOs, as they describe how "CRISPRCas9 gained rapid popularity in crop improvement programs because of its simple design, less time consumption, cost effectiveness, good reproducibility, high efficiency, precise targeting, and diverse applications" (Ahmad, Aftab, et al. 4). Due to these benefits of CRISPR, it will soon be able to replace most GMO techniques. And thanks to its lower cost and time effectiveness, it will lower the prices of food, increasing food access. CRISPR-modified food offers several advantages: it can enhance food appeal, reduce waste, improve food security, increase disease resistance, boost nutritional value, and enhance efficiency in storage and transportation (Barrell and French). In addition to being effective, CRISPR can alter the nutritional content of food. This can be vital to a growing population struggling with hunger, as providing accessible, healthy food that has a longer shelf life and durability can solve many hunger issues.

Regulatory Changes in Gene Editing

Recently, the European Parliament has finally come to a conclusion regarding new genomic techniques after waiting since 2019. Their administrative branch has decided to ease these constraints on new genomic technologies, including CRISPR, declaring that if the changes made could have been done with conventional breeding, their legislation would be inapplicable. As of fall 2025, the Genetic Technology Act will come into full effect. This act aimed to help the use of precision breeding techniques in agriculture, such as CRISPR, providing more clarity regarding its regulatory process. This would pave the way for GM products to be found in all grocery stores by 2026, cutting prices and saving lives.

Future Prospects of CRISPR-Edited Crops

CRISPR represents a transformative advancement in agricultural technology, poised to significantly enhance the nutritional quality, safety, and resilience of crops; by addressing ethical and moral concerns proactively, we can maximize the benefits of this innovation, revolutionizing food access for millions around the globe. CRISPR opens new doors for many, allowing those who have starved for days to eat. As we face the relentless challenges of climate change and wealth inequality, we must embrace forward-thinking solutions like CRISPR. While CRISPR cannot change the damages of our past, we must continue to advance and grow as a society. Just as past innovations once seemed daunting yet ultimately paved the way for progress, so too can CRISPR lead us toward a future where hunger is diminished and families are freed from the daily struggle for food. Embracing this technology could be the key to unlocking a door that leads to a healthier, more equitable world for all.

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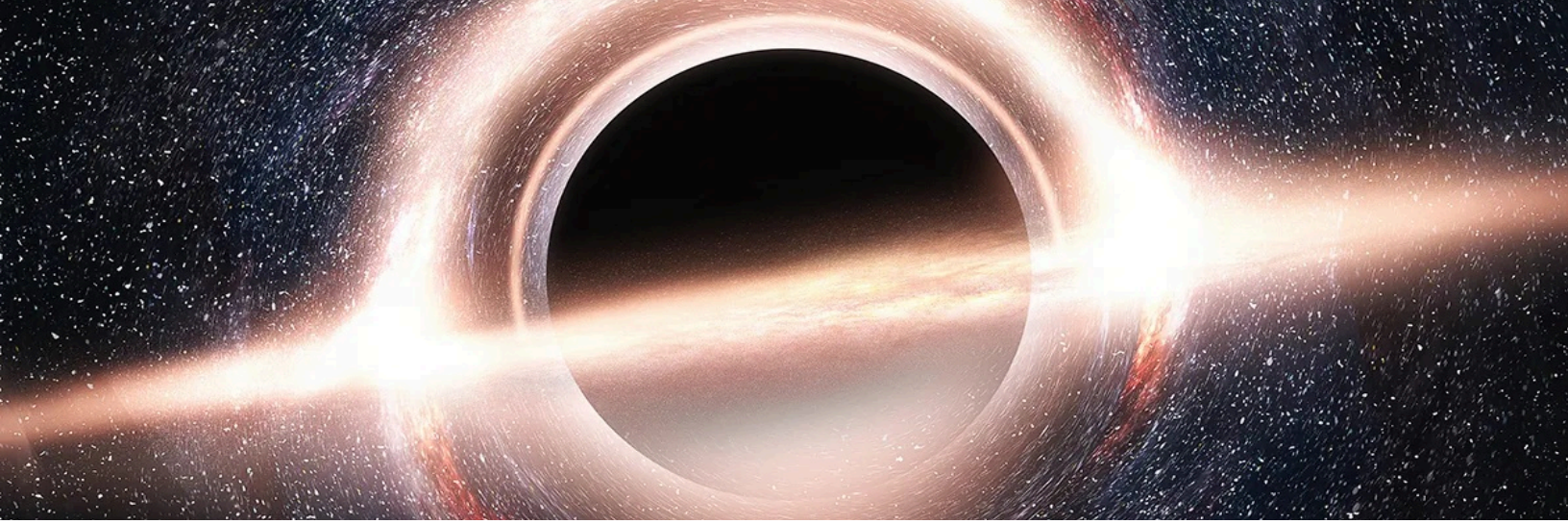

James Webb Space Telescope Discovery of Possible “Black Hole Stars” in the Early Universe

Emily Rosenthal '28

The James Webb Space Telescope (JWST) (Figure 1), launched in December 2021, has revolutionized modern astronomy by observing faint infrared light from the earliest stars and galaxies formed following the Big Bang. With its unmatched sensitivity, JWST enables scientists to look back over 13 billion years and glimpse the universe’s earliest stages of evolution (Karantha, 2025).

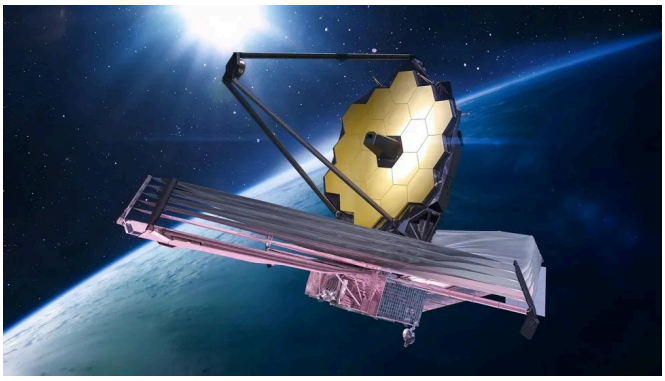

Figure 1: The James Webb Space Telescope [Image] (Karantha, 2025).

In 2025, JWST detected a set of compact, mysterious objects that defied classification as ordinary stars, galaxies, or quasars (Figure 2). These observations revealed hundreds of small, intensely red sources from the universe’s first billion years that appeared far too bright and compact to fit existing astrophysical models. In September 2025, researchers suggested that these enigmatic bodies may represent a new type of cosmic object known as “black hole stars.” These are theorized to be rapidly growing black holes enveloped in thick layers of gas and dust that absorb and re-emit radiation, making them appear deceptively similar to ordinary stars (Clery, 2025). If confirmed, these discoveries could provide crucial insight into how black holes formed and expanded rapidly in the early universe.

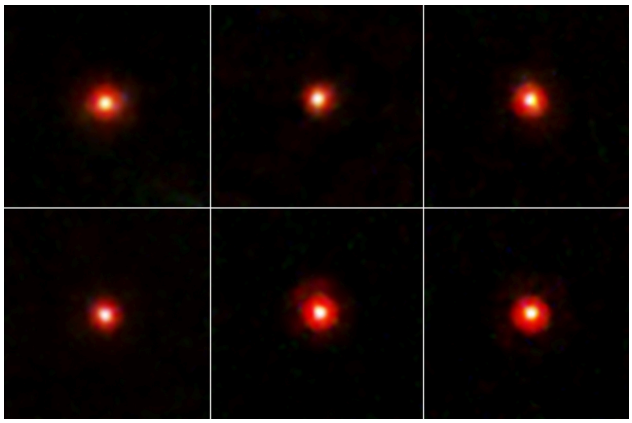

Figure 2: Hundreds of mysterious objects detected by JWST from the universe’s first billion years of existence [Image] (Clery, 2025).

How the Discovery Was Made

Using its Near Infrared Spectrograph (NIRSpec) and Near Infrared Camera (NIRCam), JWST conducted an extensive deep-space survey known as the Red Unknowns: Bright Infrared Extragalactic Survey (RUBIES). During this program, astronomers identified numerous compact red sources that were too luminous to be typical early galaxies and too small to resemble quasars.

One of these sources, officially designated RUBIES-UDS-154183 but nicknamed “The Cliff,” showed a sudden drop in its spectral light output due to hydrogen absorption, which is known as a Balmer Break (Berard, 2025) (Figure 3). While this feature occasionally indicates a population of old stars, The Cliff’s energy distribution and extreme brightness suggest that a powerful central energy source, such as a black hole, is hidden inside a thick gaseous layer.

Computer simulations reinforced this hypothesis, revealing

how when a black hole forms within a massive cloud of gas, the gas collapses inward and heats up, releasing immense amounts of radiation. Due to the extreme density of the surrounding gas, much of this energy is trapped and reprocessed, escaping only as red and infrared light (Starr, 2025). These findings, first reported in June 2025 and published in Astronomy & Astrophysics in September 2025, represent the first formal proposal for the existence of black hole stars.

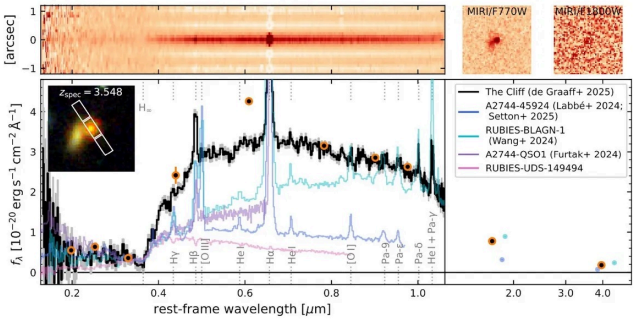

Figure 3: The drop in the light spectrum of "The Cliff" reveals a Balmer Break caused by hydrogen absorption [Graph] (Karantha, 2025)

Implications for the Early Universe

If black hole stars exist, they may represent a missing evolutionary link between the first collapsing gas clouds and the supermassive black holes that later formed. By allowing black holes to grow rapidly within dense environments, they could provide a natural explanation for how enormous black holes existed so soon after the Big Bang.

The discovery also offers a new perspective on the mysterious "little red dots" visible throughout JWST's deep-field images. They were initially assumed to be small galaxies, but their light signatures align more closely with the black hole star model. Black hole stars may have been a common feature of the early cosmos, playing a key role in cosmic reionization when neutral hydrogen atoms were ionized by radiation and the universe transformed from opaque to transparent (Berard, 2025). The proposal of black hole stars in 2025 marks a bold advancement in understanding how the universe's first massive structures formed, resolving a long-standing mystery that conventional galaxy formation theories struggled to address.

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Why is Cancer Becoming More Prominent in Young People?

Sienna Schwartz '27

Cancer, a disease where cells grow and divide uncontrollably, used to be classified as a disease of age. Yet today, doctors are seeing an unsettling shift: people in their early 20s, 30s, and 40s are being increasingly diagnosed with cancers once thought to affect mainly older adults (Lambert, 2024).

According to the National Cancer Institute (2024), rates of cancer among adults under 50 have been rising since the 1990s, particularly for colorectal, breast, thyroid, and pancreatic cancers (Phillips, 2025). While overall cancer deaths have declined due to improved detection and treatment, the increase in younger populations is defying expectations, see Figure 1. The race to identify the causes of these early onset cancers has begun.

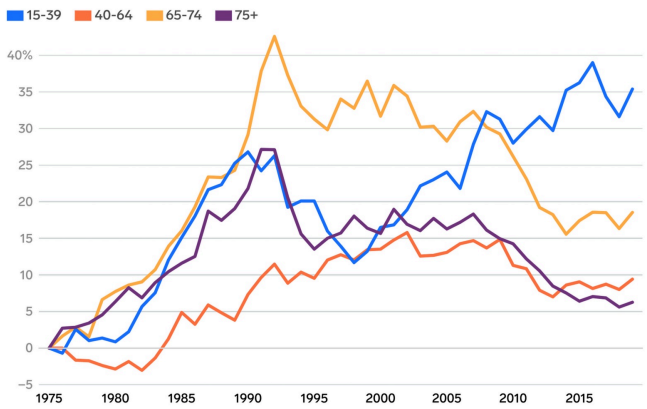

Figure 1: This figure displays the cumulative percent change in cancer incidence rates per 100,000 from 1975 by age group (Hoff, 2024).

One of the leading theories is centered around lifestyle changes. Such environmental habits include the consumption of ultra-processed foods, the rise in obesity rates, and the disruption of sleep cycles which all may play a role in fueling early tumor growth (Science News, 2024). Experts from Memorial Sloan Kettering Cancer Center (MSKCC) add that factors such as antibiotic use, chronic stress, and changes to the gut microbiome could also influence cancer risk in younger adults. "We're seeing the effects of decades of cumulative exposure to factors that promote inflammation and metabolic stress," explained Dr. Kimmie Ng, a gastrointestinal oncologist at MSKCC (Memorial Sloan Kettering Cancer Center, 2024).

However, another theory was explored in a 2013 investigation by The Guardian. It revealed that many young patients report their early symptoms being dismissed or misdiagnosed by doctors, who often assume that serious cancers are rare in younger people (Boseley, 2013). This leads to delayed diagnoses, often until the disease has reached advanced stages, leaving patients with fewer treatment options. The emotional impact of this experience has left many young adults describing themselves as a "lost generation" of cancer patients (Boseley, 2013).

The effects of childhood cancer extend far beyond the hospital. A report by Today Online (2024) found that young adults with cancer face unique emotional and financial challenges. In fact, they often struggle with identity, fertility, and career stability while coping with the demands of treatment, see Figure 2 (Loh, 2024) .

Figure 2: A woman receiving radiation therapy treatment for cancer. Emotional, physical, and financial support are crucial after a cancer diagnosis.

Despite the challenges, researchers view these circumstances as a moment for change as some progress has been made such as lowering the recommended screening age for colorectal cancer from 50 to 45 (National Cancer Institute, 2024). Raising awareness among both healthcare providers and the public is also essential to ensure symptoms in younger adults are not overlooked.

As Dr. Philip Rosenberg from the NCI notes, “Understanding why these cancers are rising could reshape how we think about prevention and screening for generations to come” (National Cancer Institute, 2024). As early-onset cancer reshapes the landscape of modern medicine, the next challenge lies in turning awareness into action, and research into prevention.

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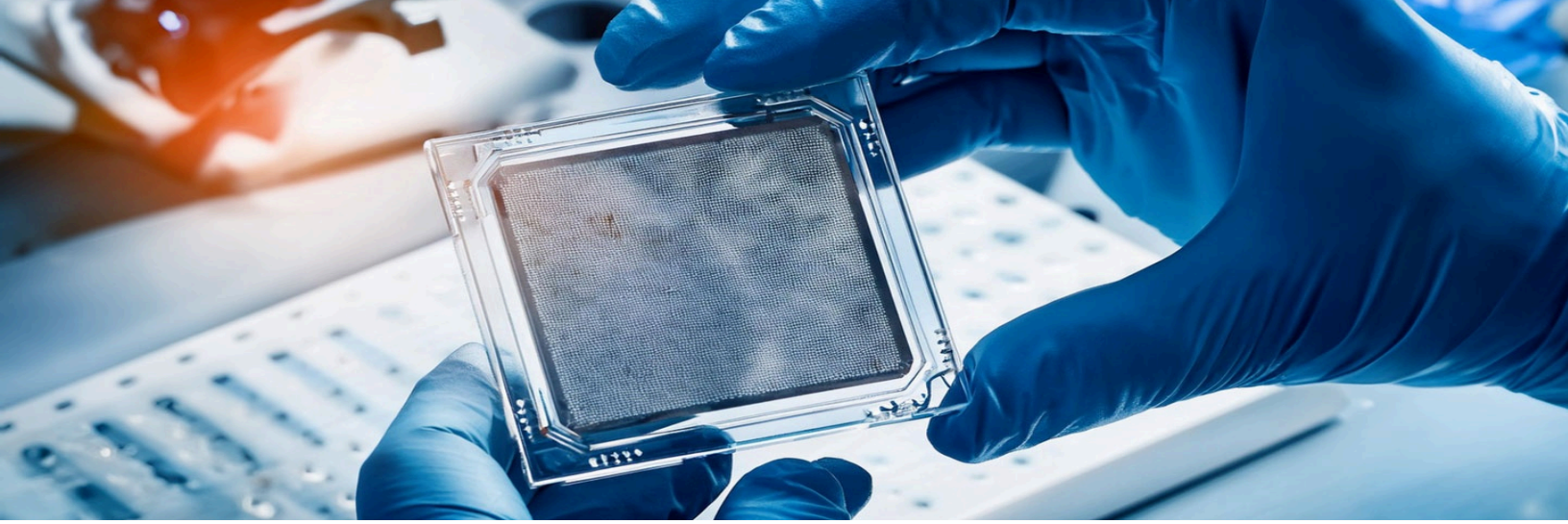

TOXBOX: The Journey to Chemical Safety

By Azita Vazhayil '28

A widely used drug in the 1950s, Thalidomide was marketed as a safe sleep aid and drug meant to help pregnant women overcome morning sickness. However, it caused severe birth defects in over 10,000 babies internationally, due to a lack of sufficient toxicology tests. Traditional testing methods, like animal testing, failed to predict human-specific toxicity. With the development of new chemicals and materials come concerns about such toxicity, which impacts health and regulatory compliance. Toxicology methods have been developed, but they fail to address toxicity's whole chemical cycle. They are also time-consuming, costly, and lack the necessary flexibility to meet the needs of modern testing. This is why the European Union (EU) developed TOXBOX.

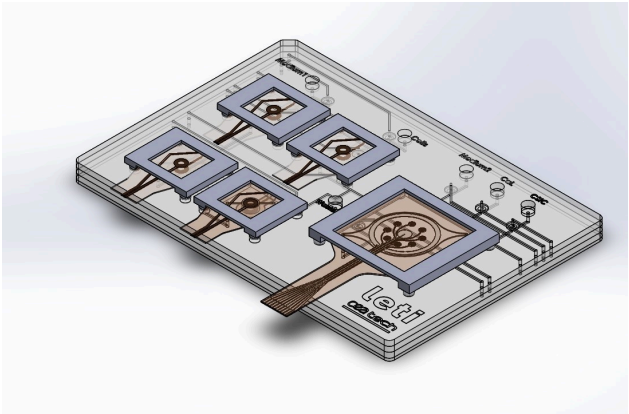

Figure 1: A design of the TOXBOX platform.

What is TOXBOX?

TOXBOX is a coherent, easy-to-use platform that helps deliver comprehensive toxicity data. It combines advanced testing methods with better data analysis, helping reduce reliance on animal testing to better meet public demands. It uses in-vitro and in-silico models, non-invasive sensors, and environmental tests to achieve its goals.

In-vitro tests involve using organoid-on-chip models, simulating human organs to realistically test toxicity, and in-silico models use AI to predict long-term chemical impacts. Environmental tests include using zebrafish embryos, data science methods, and non-invasive sensors to monitor cellular responses. Altogether, these elements form a platform powered by AI that can analyze microscopy images instantly and provide automated toxicological assessments. Its testing is highly efficient, producing large amounts of data in a short amount of time. TOXBOX was created to support the safe, sustainable development of chemicals.

Why is TOXBOX better than previous methods of testing?

TOXBOX uses comprehensive testing to help create a platform for detailed, accurate toxicity testing. It integrates advanced data and uses modular systems for environmental and human safety evaluations. Interlaboratory validation allows for reliable results by testing in different labs and life

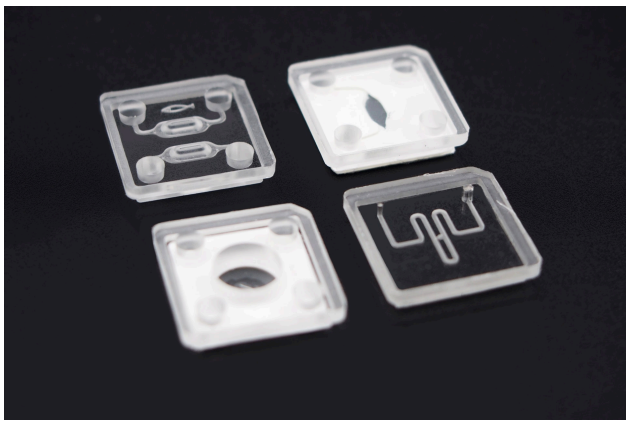

Figure 2: A design of the TOXBOX chip models.

cycle stages. TOXBOX reduces animal use by adopting these different testing methods and makes testing processes increasingly more efficient and cost-effective. It helps to facilitate international safety standards and regulations and is designed for safer and more sustainable products. Overall, TOXBOX helps solve previous issues in the toxicology field that have served as blockades for progress in chemical development.

Limitations of TOXBOX

While TOXBOX is an extremely promising project, it does have limitations. For example, it is still in its development phase; the project's funding is extended to 2027, implying that it still has room to develop for the next two years (Nanobiotechnology For Diagnostics & Consejo Superior Con Investegaciones Cientificas, 2023). This serves as an issue, as TOXBOX cannot fully go into effect until its completion. As companies continue developing new products, the harmful and costly methods used today may become more common. Additionally, a goal of the project is to create a "plug and play" system that can be used across laboratories worldwide (CORDIS, 2025). This may be a severe undertaking of what the project will actually be able to do, which could affect funding for its development in the future. Therefore, TOXBOX still has hurdles it must overcome before it can be implemented worldwide.

The Future of TOXBOX

TOXBOX has a bright and promising future ahead of it. Since its beginning, it has developed prototype chips and microfluidic architectures for organ-on-a-chip systems and created transgenic zebrafish and organoid models. TOXBOX is also currently publishing new environmental testing methods and establishing collaborations with various EU projects and the EU Joint Research Center. This project will continue until December 2027, hoping to achieve its goal of advancing innovation in chemicals and materials safely across Europe.

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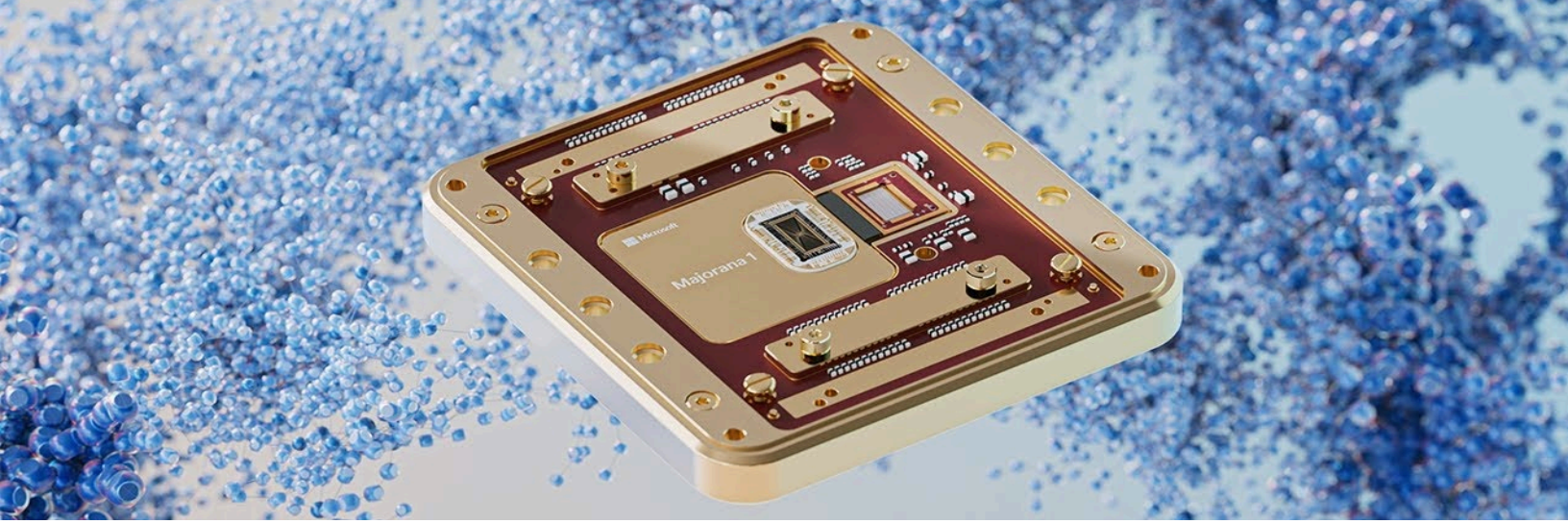

Can a company hold all of the world's computational power in a single chip?

Anita Vinokur, '29

Quantum computing, overall, can revolutionize technology, promising to help solve problems that even the most powerful supercomputers cannot solve. Quantum computers can reinvent how new life-saving drugs are made and come up with new approaches to solving climate change. They are not only powerful, but operate much faster than any computer or supercomputer made today. But if quantum computers are so revolutionary, why are quantum computers, or chips, not used today?

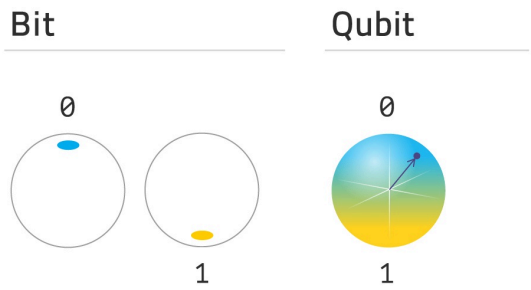

Figure 1: This is the difference between a bit, where the value can be either 0 or 1, and a qubit that can be both at the same time. The qubit is in superposition as shown in the image.

In order to understand why they are not as prominent, it is important to know how a quantum computer works. Regular computers use bits (binary digits), which hold one value, either 0s or 1s. Quantum computers use qubits, the “basic unit of information in quantum computers” (Scheinder, 2024) that can exist as 0, 1, or both simultaneously; this state is known as superposition. (Figure 1) This property, along with entanglement, “ a non-classical correlation, or shared quantum state, between two or more quantum systems (or quantum particles) even if they are separated by a large distance”. Microsoft allows quantum computers to perform a massive number of calculations in parallel.

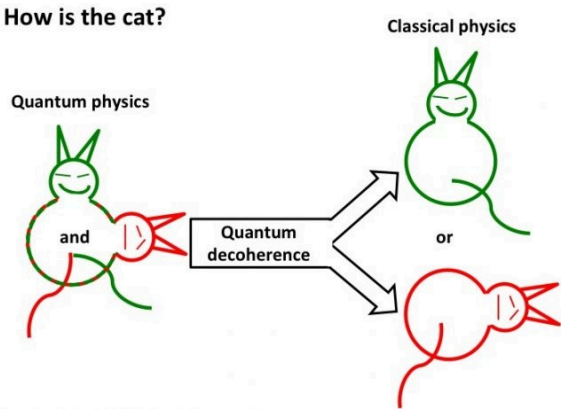

Figure 2: Schrodinger's cat, and an example of how when a qubit is in superposition, it can show the cat being dead or alive, but after decoherence, the qubit can only have one value, either dead or alive.

However, quantum computers do have downsides. Qubits are susceptible to external noise, most often in the form of vibrations, heat, and external electromagnetic fields; interference of any magnitude can alter a qubits phase, making qubits "notoriously finicky and prone to errors" (Orf, 2024). If a qubit is disturbed, it loses its quantum relative phase and as a result, its state, causing the qubit to turn into a regular bit through a process called decoherence. To understand decoherence, one can imagine Schrodinger’s cat, which is the idea that if there is a 50% chance that a cat is dead or alive in a box; we don't know if the cat is dead or alive until we check, meaning the cat is dead and alive at the same time. A qubit will store the value of the cat as both dead and alive. After decoherence, the qubit can only store one value, that the cat is either dead or alive, not both. (Figure 2) Because of the fragility of a qubit, it is very easy for a quantum computer to

become a regular computer. (National Science Foundation, 2025)

For several years, this problem has stumped numerous researchers. Everyone strived to make a quantum chip but struggled to maintain a stable environment free from noise. Unlike many of these research facilities, and scientists, Microsoft looked for the answer in material science rather than chip design. Through decades of research, Microsoft discovered the topological superconductor (topoconductor). A topoconductor is a unique state of matter combining the properties of a superconductive material (a material that can conduct electricity with near zero resistance) and those of topology. The qubits in the chips are made for a combination of indium arsenide,a semiconductor, and aluminum,a superconductor, to create the new state of matter: the topoconductor Figuree 3), which allows information to be stored, measured, and recorded digitally.

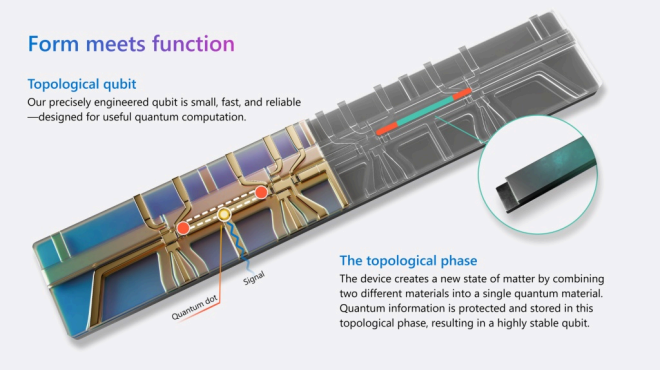

Figure 3: This image shows the qubit and the topological phase in the qubit itself.

Microsoft utilizes the superconductor to store Majorana fermions or Majorana Zero Modes (MZMs). MZMs are exotic particles that can create an ultra stable quantum state and the robust edge state, where "electrons can flow without friction, gliding effortlessly around obstacles as they stick to their perimeter-focused flow" (Chu, 2024). This property permits quantum information to be stored in an extremely stable and noise-resistant state allowed by the topoconductor, MZMs and robust edge state. This makes this quint more reliable because they store information in a topological e spread across the MZMs instead of in one fragile physical property. Imagine that the qubit is stored in almost another dimension that can't be reached by the outside noise and heat.

Microsoft's Majorana 1 chip is the first quantum processor to use a topological core, storing information in stable Majorana-based qubits. Although the chip currently has only eight qubits, it is engineered to scale to one million qubits on a single chip, something impossible with traditional designs. This scalability comes from the natural stability of topological

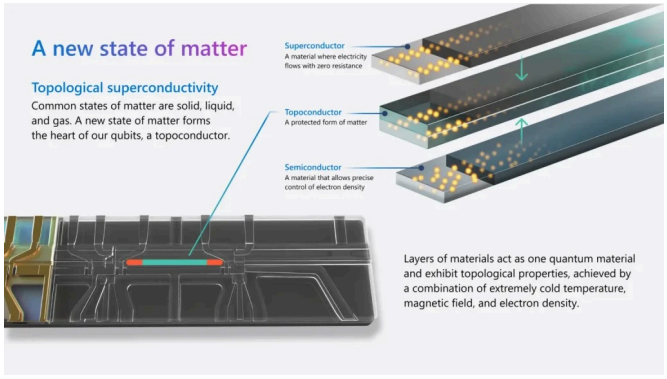

Figure 4: This image is an explanation of what is inside the qubits. It shows the combination of the materials to make the topological topoconductor and how they form topological superconductivity.

qubits, which need far less error correction than normal qubits. Microsoft plans to use this approach to build a "fault-silent prototype" in the coming decades, a major step toward a practical, large-scale quantum computer.

The Mojarana 1 currently only has 8 qubits, but Microsoft has stated that the chip is designed to scale " to a million qubits on a single chip." (Nayak, 2025). Using the topoconductor, Microsoft is planning on making a fully operational prototype that they call a fault silent prototype (FTP) "not in years but decades"(Microsoft, 2025). Once they build it, the new technology could solve problems that regular computers would take centuries to solve.

The new FTP built with the topoconductor would be able to do many things not thought of as possible a few years ago. Eventually, this advancement in quantum computing could find solutions to self-repairing materials for cracked roads and bridges. It could also find cures for cancer, HIV, and many other illnesses and diseases for all people. Additionally, it could perfectly calculate hard problems in milliseconds, unlike regular computers that could take billions of years. Take the traveling salesperson problem (TSP). TSP is a well-known difficult problem in computer science. It asks for the shortest amount of distance one will travel amongst given cities, if each city must be visited exactly once. As the number of cities increases, it becomes increasingly difficult for a computer to calculate this; it will take so much time to finish that the whole universe could explode before the computer finishes. But this new quantum chip can make that calculation in seconds.

Despite Microsoft's promising results, some scientists question whether the company has truly demonstrated topological qubits. Some point out that the data Microsoft released is limited and that several of the observed signals could also be produced by non-topological states. Because the research has not yet been independently replicated at scale, many experts remain cautious about calling these results

definitive. This skepticism highlights that, while the Majorana 1 chip is an exciting development, the underlying physics still needs stronger evidence before the claims can be fully accepted.

In conclusion, Microsoft's Majorana 1 chip is one of the most important steps technology has made toward making quantum computing stable, scalable, and practical. By using topological qubits and Majorana particles to build topological qubits, Microsoft has solved one of the biggest problems in quantum technology: how to keep qubits stable and not interfere with outside noise. The creation of a topological state of matter will start a new era in technology and quantum computing as a whole. If Microsoft can scale the chip to over 1 million qubits, it will transform how people solve problems. From discovering new medicine to predicting climate change, the possibilities of problems that this innovation can solve are endless. The Majarana 1 chip displays how far scientists have come with quantum computing, turning the theory into reality.

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NASA's Perseverance Rover Locates Evidence to Past Microbial Life on Mars

Jessica Willis '27

In July 2024, NASA's Perseverance rover made a discovery while exploring the ancient riverbed in Jezero Crater on Mars. The rover has been studying many igneous and sedimentary rocks in the crater, looking for early Martian biosignatures and hints to geologic processes. A potential biosignature is a structure or substance that may have a biological background, but lacks data to make conclusions about the presence of life. On the crater's western edge, Perseverance came across a rock nicknamed "Cheyava Falls" in the Bright Angel formation. Bright Angel is a set of rocky mudstone and conglomerate outcrops on the northern and southern edges of Neretva Vallis (Taveau, 2025). Neretva Vallis is an ancient river channel that flowed through Jezero Crater a long time ago. On the 10th of September, 2025, the NASA's Perseverance team announced that Cheyava Falls holds possible biosignatures (NASA, 2025). The rock appeared to be covered in tiny dark spots and larger circles. This surprised scientists on Earth because these types of markings are often linked to microbes. If this was seen on Earth, geologists could conclude that those markings are most likely biological, but given the location of this sample, it raised questions. These spots seem as if they are hints to life, having been fossilized in the Martian mud. The rover's astrobiology mission is to search for signs of past microbial life, characterizing Mars geology and collecting samples for the future return to Earth

The Jezero crater was formed by an asteroid impact roughly 4 billion years ago. After Perseverance's landing in February 2021, the mission team discovered that the crater's floor is composed of igneous rock that had formed by volcanic activity on the surface and/or magma underground. It is located in the Syrtis Major quadrangle, also known as MC-13. MC-13 is a series of 30 maps of locations on Mars, covering longitudes 270° to 315° west and latitudes 0° to 30° north (NASA, 1998). Syrtis Major has several types of igneous rocks such as basalt, dacite and granite, while also containing the mineral olivine in some areas in the quadrangle. Because olivine alters into other minerals when exposed to water, both the significant abundance and distribution throughout the rocks are notable. Images taken from Mars reconnaissance orbiter (MRO) shows a river delta, Neretva Vallis. This is not only clear evidence to a pre-existing lake, but also delta rocks that are prime environments for entombing signs of ancient life in fossil form (NASA, 2023). MRO gathered mineral data from two of its instruments, those being the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) and the Context Camera (CTX) (NASA, 2024a). This data is then overlaid on an image of Perseverances landing site to show the locations of certain minerals hidden in the Martian rock (see Figure 1).

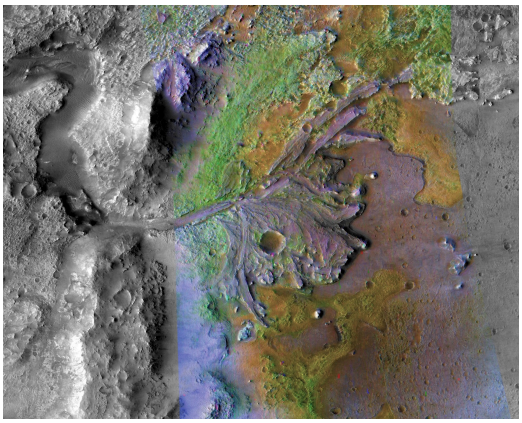

Figure 1: Image of Jezero Crater overlaid with mineral data gathered from Mars orbiter. Green represents carbonates; minerals that form in wet environments favorable to preserving ancient life (NASA, 2024a).

Along the northern wall of Neretva Vallis channel is the Bright Angel outcrop, its brighter tone indicates significantly

older rock exposed by river erosion. In July of 2024, Perseverance imaged a feature within this rock known as Cheyava Falls on its 1,212th Martian day of the mission, which measures approximately 3.2 feet by 2 feet. The image was captured using WATSON, a high resolution topographical sensor mounted on the SHERLOC instrument on Perseverance's 7 foot robotic arm (NASA-JPL, 2024). WATSON is capable of taking high-resolution, close up images of rock textures. SHERLOC uses a laser and

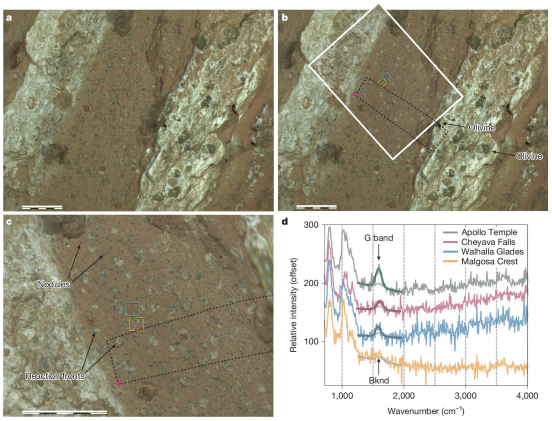

Figure 2: a, Image of Cheyava Falls captured using WATSON. b, Image of Cheyava Falls with annotations pointing to Olivine in the rock. c, Image of Cheyava Falls with annotations showing the footprint of the PIXL scan and the authigenic nodule. d, data collected by Perseverance's 'Scanning Habitable Environments with Raman & Luminescence for Organics & Chemicals' (SHERLOC), a spectrometer from the Bright Angel rock formation, showing Raman spectroscopy readings. One line being Cheyava Falls (second line from top), displaying "G-band" signals consistent with organic molecules. "Bknd" is short for background (Hurowitz et al., 2024).

spectroscopy to identify such organic molecules and minerals within rocks, detecting carbon based molecules. The rock is reddish tan colored, its layers separated by thinner light-toned layers that strike out at an angle, signifying its displacement from its original orientation, pointing to geological activity in the area (see Figure 2a). Additionally, the brighter tone of the outcrop indicates that these rocks have gone through long periods of exposure and weathering compared to the surrounding material, providing a window into much older geological history. SHERLOC and PIXL highlight olivine as an authigenic nodule, meaning it formed where it is located (see Figure 2b-c). On Earth, organic material in sedimentary rocks form in aqueous environments, offering a favorable environment for life to exist. The combination of water erosion and detected organics provides probable, although not definitive, indication of past life. In this context, identifying the carbon as either organic or nonorganic is an important task because carbon is a key chemical building block for life, making it an important target.

These characteristics, though offering highly plausible evidence, do not constitute exact proof of biological activity. Therefore, other possible explanations must be explored. A variety of organic carbon compounds are known to support the non-biological (abiotic) process known as the reductive dissolution of ferric iron oxide minerals in certain temperatures (Hurowitz et al., 2024). This means that certain organic molecules can chemically react with iron materials, without involving life. Abiotic organics can emit Raman signals that can resemble those made by biological carbon, making it more difficult to distinguish between the two circumstances (see Figure 2d). While carbon is necessary for life, its presence alone is not sufficient evidence as extraterrestrial processes, like asteroid impacts, can produce similar signatures. The presence of organic matter could have been produced through abiotic synthesis or delivered from non-biological sources, such as volcanic activity or meteorites, which implies that these such reactions could have occurred (see Figure 2d) (Hurowitz et al., 2024). While this is possible on Earth, further analysis is required to confirm if the organic materials in the martian rocks can drive this process. Only through additional contextual data and the return of the rock samples can these alternative conclusions be confirmed. Cheyava Falls is the 25th sample of what is currently 30 samples taken from Mars. Perseverance has been drilling into the ground at varied locations to ensure scientific diversity and redundancy. Most samples are stored within Perseverance, while there are 10 out of the 43 eventual samples left at the sample depot as backups, located in the Three Fork region in Jezero crater (NASA, 2024b). This was created as a backup plan in case the samples stored on the rover were unable to be collected and returned to Earth. This backup depot demonstrates NASAs commitment to ensuring that at least a portion of the samples are recoverable in the event of a hardware failure or an unplanned end to the mission. All samples are to be returned to Earth for an in-depth study. Returning the samples to laboratories on Earth will ultimately provide the best opportunity to study and understand the processes and makeup of these rocks that gave rise to such physical and molecular features. While Perseverance does carry high tech equipment like SHERLOC and PIXL, those instruments can only provide the preliminary data measurements. Only through Earth-based labs can scientists perform higher precision analysis of the rocks. NASA and the European Space Agency (ESA) are developing plans for the return of the samples (Carney, 2025a). However, as of right now, there is no confirmed date for the sample recovery and therefore can draw the prediction that if it does happen, it will not be within this decade.

The impact of a deep study of these samples would be revolutionizing. It would offer a new understanding of biology and the evolution of our universe, with the suggestion that life could be more widespread and common than previously thought. This discovery would directly inform how life evolves and persists even under harsh conditions such as Mars, having the possibility to redefine the conditions that life habitability is possible. And of course it may raise debates regarding future human travel to Mars. More specifically, concerns would arise regarding human contamination of the possible microbial life on Mars. And not only the rights humans have to colonize the planet, but also if it is necessary. This extends into ethical questions such as whether or not future Human settlement should proceed if there is indigenous life on Mars, and whether that life warrants complete isolation and/or protection. On a broader scale, the confirmation of life on Mars would have major implications towards the Great Filter hypothesis. As it stands, the theory suggests that the filter, a series of challenges that are extremely difficult for life to surpass, is behind us and humanity has overcome that filter (Re, 2022). If the signatures on Mars are early traces of life, then that suggests that the hardship has yet to come if life persisted twice. Consequently this would raise concerns and questions about the long term survivability of advanced civilizations, and the future of Humanity.

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